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On the realized joint Laplace transform of volatilities with application to test the volatility dependence

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In this paper, we investigate the estimation of the empirical joint Laplace transform of volatilities of two semi-martingales within a fixed time interval $[0, T]$ by using overlapped increments of high-frequency data. The proposed estimator is robust to the presence of finite variation jumps in price processes. The related functional central limit theorem for the proposed estimator has been established. Compared with the estimator with non-overlapped increments, the estimator with overlapped increments improves the asymptotic estimation efficiency. Furthermore, we study the asymptotic theory of estimator under the long time span setting and employ it to create a feasible test for the dependence between volatilities. Finally, simulation and empirical studies demonstrate the performance of proposed estimators.

Keywords: Itô semi-martingale; High-frequency data; Realized joint Laplace transform of volatility; Stable convergence; Volatility dependence

1. Introduction

In recent years, blooming global commerce and modern digital technologies have produced substantial volumes of data. Nevertheless, analyzing large-scale, high-dimensional datasets remains challenging. With the wide availability of high-frequency data, research on financial econometrics has been rapidly developed. Among others, volatility estimation is one of the most popular topics due to its widespread use in option pricing, risk management, and high-frequency trading, see Barndorff-Nielsen and Shephard (2002), Jacod (2008) and Andersen *et al.* (2010), etc. Statistical inference for the volatility is inherently complex since most of the studies assume that the asset prices follow Itô semi-martingale and the underlying volatility process is latent, see Merton (1973), Heston (1993), Bates (1996), Andersen *et al.* (2001b), Tijms (2003) and the references therein. Fortunately, the recent accessibility of high-frequency data has made it feasible to estimate the volatility of the price process within a fixed period. Under the assumption of continuous time models, there are usually two ways to define volatility. The first is called spot volatility, which refers to the value of the coefficient that describes the diffusion process of price return fluctuating at a specific point in time. Another type of volatility, integrated volatility, is the integration of spot volatility within a period (such as a day). Spot volatility and integrated volatility have been widely studied using high frequency data, see, e.g. Andersen *et al.* (2001b), Barndorff-Nielsen and Shephard (2002), Fan and Wang (2008), Jacod (2008), Jacod *et al.* (2009), Mancini (2009), etc.

The volatility process is random. Therefore, besides the statistical inference of volatility itself, how to use high-frequency data to obtain more information about volatility, such as the distributional pattern of volatility, has become a concern for statisticians and financial economists. Todorov and Tauchen (2012a) first proposed the realized Laplace transform (RLT) of volatility, which is a consistent estimator of the empirical Laplace transform (ELT) of volatility. The empirical Laplace transform of volatility contains more information than the integrated volatility. For instance, the empirical Laplace transform of volatility is a mapping from the data to a random function, while the integrated volatility is only a mapping from the data to a random variable. Hence, the integrated volatility can be obtained from the empirical Laplace transform of volatility. Moreover, the empirical Laplace transform of volatility preserves information about the characteristics of volatility distribution under

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some mild stationarity conditions. Shortly thereafter, Todorov and Tauchen (2012b) extended the theory of realized Laplace transform of volatility to the pure-jump model setting. Wang *et al.* (2019a) and Wang *et al.* (2019b) developed the limiting behavior of realized Laplace transform of volatility with the presence of microstructure noise. Hounyo *et al.* (2023) studied the bootstrap inference of the realized Laplace transform of volatility and applied to the bootstrap inference of the spot volatility measures. The large deviation principle of the realized Laplace transform of volatility has been derived by Feng *et al.* (2022).

Besides the Laplace transform, the distributional inference of volatility can also be based on the realized distribution of volatility directly. In this direction, Li *et al.* (2013) studied the estimation of volatility occupation time and derived the order of estimation error, and Li *et al.* (2016) further studied the volatility occupation time using the Laplace inversion. Zhang *et al.* (2022) extended the result of Li *et al.* (2013) to occupation density of volatility. Christensen *et al.* (2019) studied the realized empirical distribution of volatility in the long time span setting and applied to the goodness-of-fit test for volatility.

In this paper, we are interested in the distributional inference of a bivariate volatility process. Besides the dependence among asset prices, the dependent structure of volatility of price assets is also important in risk management, portfolio allocation, etc. With the prosperous development of financial industries, many financial innovations involve complex derivations and structured financial products, such as Collateralized Debt Obligation (CDOs). The past several years have witnessed an increasing number of research focusing on the dependent structure of two or more volatilities. For example, Andersen *et al.* (2001a) and Andersen *et al.* (2001b) provided a comprehensive analysis on the relationship between volatility and volatility, between volatility and correlation, and between correlation and correlation, for the stock return volatility and exchange rate volatility, respectively, where they found that the multivariate volatility exhibit strong one-factor structure, which is consistent with the result of Ding *et al.* (2025), the model of Ding *et al.* (2025) is very helpful in modeling the co-movement of multiple volatilities. Moreover, Kong (2017) discussed the potential volatility factor structures, Hounyo *et al.* (2023) considered bootstrap inference of spot covariance and correlation.

In this paper, rather than pursuing a more accurate dynamic model for the multivariate volatility, we aim to propose a concrete measure for the dependence of volatilities, so that it enable to understand the joint distribution of multivariate volatilities, hence can be applied to analyze the relationship among the volatilities of different assets. Precisely, we aim to determine whether the distribution of volatility of two assets in different sectors or the same sector across different time periods exhibits similarity and we seek to ascertain whether the volatility of different assets is independent and so on. For this purpose, the measure we proposed is called the empirical joint Laplace transform of volatilities, defined as

$$\int_0^T \mathrm{e}^{-\langle (u,v),((\sigma_s^X)^2,(\sigma_s^Y)^2)\rangle}\mathrm{d}s, \tag{1}$$

where $(\sigma^X)^2$ and $(\sigma^Y)^2$ are the two volatility processes of two assets X and Y, respectively. Recall that the joint Laplace transform contains full information on the distribution of a bivariate random vector. Hence, it can be used to detect the dependence of two random variables. For instance, if we consider a long enough time interval $[0,T]$, then under some appropriate stationarity and mixing conditions, as stated in DeHay (2005), the empirical joint Laplace transform $\frac{1}{T}\int_0^T \mathrm{e}^{-\langle (u,v),((\sigma_s^X)^2,(\sigma_s^Y)^2)\rangle}\mathrm{d}s$ will be close to the joint Laplace transform of $(\sigma_t^X)^2$ and $(\sigma_t^Y)^2$, namely, $\mathbb{E}[e^{-u(\sigma_t^X)^2-v(\sigma_t^Y)^2}]$. It thus can be used to either recover the joint distribution of two volatilities, study their independence, or investigate any other quantities related to the joint distribution. Since the empirical joint Laplace transform $\int_0^T \mathrm{e}^{-\langle (u,v),((\sigma_s^X)^2,(\sigma_s^Y)^2)\rangle}\mathrm{d}s$ is unobservable, hence we will first construct a consistent estimator for fixed time interval $[0,T]$ by using high-frequency data and then we derive the asymptotic behavior under the long time span.

The rest of this paper is organized as follows. In section 2, we introduce the model assumptions and present the estimators. Section 3 derives the consistency and central limit theorem of the proposed estimators for fixed time interval $[0,T]$. The asymptotic behavior of the estimator under a long time period is presented in section 4. In sections 5 and 6, we exhibit the simulation results and empirical study, respectively. We conclude this paper in section 7, and all technical proofs are put into the appendix.

2. Setup

We consider a two-dimensional process $\{(X_t,Y_t), 0\le t\le T\}$, which is defined on an appropriate filtered probability space $(\Omega,\mathcal{F},(\mathcal{F}_t)_{t\ge0},\mathbb{P})$ with the following form:

$$\begin{cases} \mathrm{d}X_t = b_t^X\mathrm{d}t + \sigma_t^X\mathrm{d}W_t^X \\ \qquad + \int_{\mathbb{R}} \delta^X(t-,x)\mu^X(\mathrm{d}t,\mathrm{d}x), \quad X_0 = x, \\ \mathrm{d}Y_t = b_t^Y\mathrm{d}t + \sigma_t^Y\mathrm{d}W_t^Y \\ \qquad + \int_{\mathbb{R}} \delta^Y(t-,y)\mu^Y(\mathrm{d}t,\mathrm{d}y), \quad Y_0 = y, \end{cases}$$

where b_t^Z and σ_t^Z, $Z = X, Y$ are adapted and locally bounded càdlàg processes, (W_t^X, W_t^Y) is a two-dimensional Gaussian process with independent increments, zero mean, and the covariance matrix is

$$\begin{bmatrix} t & \int_0^t \rho_s\mathrm{d}s \\ \int_0^t \rho_s\mathrm{d}s & t \end{bmatrix}, \quad 0\le t\le T,$$

where $\rho_t, 0\le t\le T$, is a deterministic function and take values in the interval $[-1,1]$. There exists a standard Brownian motion $W_t^\star$ independent of W_t^X, we can rewrite

$$\mathrm{d}W_t^Y = \rho_t\mathrm{d}W_t^X + \sqrt{1-\rho_t^2}\mathrm{d}W_t^\star, \quad 0\le t\le T.$$

Besides, μ^X and μ^Y are homogeneous Poisson random measures with compensators $\nu^X(\mathrm{d}x)\otimes \mathrm{d}t$ and $\nu^Y(\mathrm{d}y)\otimes \mathrm{d}t$ respectively, and $\delta^X(t,x)$ and $\delta^Y(t,y)$: $\mathbb{R}^+\times\mathbb{R}\to\mathbb{R}$ are cadlag in t.

Due to technical reasons, we need two assumptions, which are similar to Todorov and Tauchen (2012a). Assumption 2.1 is used to restrict the discontinuous part (jumps), and assumption 2.2 is about the coefficient of the continuous part of price processes.

Assumption 2.1 The *Lévy* measure of μ^Z, $Z=X,Y$ satisfy

$$\mathbb{E}\left(\int_0^t\int_{\mathbb{R}}\left(|\delta^Z(s,z)|^p\vee|\delta^Z(s,z)|\right)\nu^Z(\mathrm{d}z)\mathrm{d}s\right)<\infty,$$

for every $t>0$ and every $p\in(\beta,1)$, where $0\le\beta<1$ is some constant.

Assumption 2.1 requires the jump components of price processes to be of finite variation, and the first moment of the jump processes exists. This condition is needed to show the asymptotic normality.

Assumption 2.2 Assume that σ_t^Z, $Z=X,Y$, is an Itô semimartingale given by

$$\sigma_t^Z=\sigma_0^Z+\int_0^t\tilde{b}_s^Z\mathrm{d}s+\int_0^t v_s^Z\mathrm{d}W_s^Z+\int_0^t v_s^{'Z}\mathrm{d}W_s^{'Z}+\int_0^t\int_{\mathbb{R}}\delta^{'Z}(s-,z)\mu^{'Z}(\mathrm{d}s,\mathrm{d}z),$$

where $W_s^{'Z}$ is a Brownian motion that is independent W_s^Z, $\mu^{'Z}$ is a homogenous Poisson measure with *Lévy* measure $\nu^{'Z}(\mathrm{d}z)\otimes\mathrm{d}t$, which has arbitrary dependence with μ^Z, and $\delta^{'Z}(t,z):\mathbb{R}^+\times\mathbb{R}\to\mathbb{R}$ is càdlàg in t. We have for every t and s and some $\iota>0$,

$$\mathbb{E}\left(|b_t^Z|^{3+\iota}+|\tilde{b}_t^Z|^2+|\sigma_t^Z|^2+|v_t^Z|^{3+\iota}+|v_t^{'Z}|^{3+\iota}+\int_{\mathbb{R}}|\delta^{'Z}(t,z)|^{3+\iota}\nu^{'Z}(\mathrm{d}z)\right)<C,$$

$$\mathbb{E}\left(|b_t^Z-b_s^Z|^2+|v_t^Z-v_s^Z|^2+|v_t^{'Z}-v_s^{'Z}|^2+\int_{\mathbb{R}}(\delta^{'Z}(t,z)-\delta^{'Z}(s,z))^2\nu^{'Z}(\mathrm{d}z)\right)<C|t-s|,$$

where $C>0$ is some constant that does not depend on t and s.

Assumption 2.2 imposes integrability conditions on b_t^Z and σ_t^Z, $Z=X,Y$ and bound their fluctuations over a short period of time. Moreover, we can treat b_t^Z and σ_t^Z, $Z=X,Y$ 'locally' as constants when the sampling frequency is high enough. This restriction is mild and widely used in the models of financial econometrics.

Next, we present our estimators. For a fixed time interval $[0,T]$, we observe $\{(X_t,Y_t)\}$ at discrete time points $t_i^n=i\Delta_n$. When $\Delta_n\to0$, we obtain the high frequency data $\{(X_{t_i^n},Y_{t_i^n})\}$. The quantity of interest is the empirical joint Laplace transform (integrated over the interval $[0,T]$) of σ^X and σ^Y:

$$\int_0^T \mathrm{e}^{-\langle(u,v),((\sigma_s^X)^2,(\sigma_s^Y)^2)\rangle}\mathrm{d}s,$$

for $(u,v)\in\mathbb{R}_+^2$. Let $\Delta_i^nZ:=Z_{t_i^n}-Z_{t_{i-1}^n}$ for $Z=X,Y$, and

$$\xi_i^n(u,v)=\cos\left(\frac{\sqrt{2u}\Delta_i^nX+\sqrt{2v}\Delta_{i+1}^nY}{\sqrt{\Delta_n}}\right).$$

We first propose an estimator with non-overlapped increments:

$$V_n(u,v)=2\Delta_n\sum_{i=1}^{\lfloor n/2\rfloor}\xi_{2i-1}^n(u,v),$$

where $n=\lfloor T/\Delta_n\rfloor$. Our second estimator with overlapped increments is defined as

$$U_n(u,v)=\Delta_n\sum_{i=1}^{n-1}\xi_i^n(u,v).$$

Note that in the estimator $U_n(u,v)$, two adjacent items $\xi_i^n(u,v)$ and $\xi_{i+1}^n(u,v)$ are based on two overlapping blocks, $[t_{i-1}^n,t_{i+1}^n]$, $[t_i^n,t_{i+2}^n]$, hence, we call it overlapped estimator. Under the same logic, we call $V_n(u,v)$ the non-overlapped estimator.

3. Asymptotic results

In this section, we first show the consistency of these two estimators. Let $\sigma_i^Z:=\sigma_{t_i^n-}^Z$, $\Delta_i^nW^Z:=W_{t_i^n}^Z-W_{t_{i-1}^n}^Z$, $Z=X,Y$, $\mathcal{F}_i=\mathcal{F}_{t_i^n}$. For the estimator $V_n(u,v)$, we have the following approximation:

$$\cos\left(\frac{\sqrt{2u}\Delta_{2i-1}^nX+\sqrt{2v}\Delta_{2i}^nY}{\sqrt{\Delta_n}}\right)\approx\cos\left(\frac{\sqrt{2u}\sigma_{2i-2}^X\Delta_{2i-1}^nW^X+\sqrt{2v}\sigma_{2i-2}^Y\Delta_{2i}^nW^Y}{\sqrt{\Delta_n}}\right),$$

where $i=1,\dots,\lfloor n/2\rfloor$. Using a standard argument in the high frequency statistics, we can show that this approximation error is asymptotically negligible. Moreover,

$$\begin{aligned}\mathbb{E}&\left[\cos\left(\frac{\sqrt{2u}\sigma_{2i-2}^X\Delta_{2i-1}^nW^X+\sqrt{2v}\sigma_{2i-2}^Y\Delta_{2i}^nW^Y}{\sqrt{\Delta_n}}\right)\Bigg|\mathcal{F}_{2i-2}\right]\\&=\mathbb{E}\left[\cos\left(\frac{\sqrt{2u}\sigma_{2i-2}^X\Delta_{2i-1}^nW^X}{\sqrt{\Delta_n}}\right)\cos\left(\frac{\sqrt{2v}\sigma_{2i-2}^Y\Delta_{2i}^nW^Y}{\sqrt{\Delta_n}}\right)\Bigg|\mathcal{F}_{2i-2}\right]\\&=e^{-\langle(u,v),((\sigma_{2i-2}^X)^2,(\sigma_{2i-2}^Y)^2)\rangle}.\end{aligned}\tag{2}$$

The equalities hold because $\sin(\cdot)$ is an odd function and $\Delta_{2i-1}^n W^X$ is independent of $\Delta_{2i}^n W^Y$. According to the Weak Law of Large Numbers, we have

$$2\Delta_n \sum_{i=1}^{\lfloor n/2 \rfloor} \left\{ \cos\left(\frac{\sqrt{2u}\sigma_{2i-2}^X \Delta_{2i-1}^n W^X + \sqrt{2v}\sigma_{2i-2}^Y \Delta_{2i}^n W^Y}{\sqrt{\Delta_n}} \right) - e^{-\langle (u,v),((\sigma_{2i-2}^X)^2,(\sigma_{2i-2}^Y)^2)\rangle} \right\} \xrightarrow{P} 0. \tag{3}$$

Hence, as $n \to \infty$, $2\Delta_n \sum_{i=1}^{\lfloor n/2 \rfloor} e^{-\langle (u,v),((\sigma_{2i-2}^X)^2,(\sigma_{2i-2}^Y)^2)\rangle}$ converges to $\int_0^T \mathrm{e}^{-\langle (u,v),((\sigma_s^X)^2,(\sigma_s^Y)^2)\rangle} \mathrm{d}s$. The consistency of U_n can be obtained in the same way.

Next, we present the central limit theorems of the estimators $V_n(u,v)$ and $U_n(u,v)$. The notation $\xrightarrow{\mathcal{S}}$ represents the stable convergence in law. The stable convergence in law is stronger than the convergence in law and weaker than the convergence in probability. More details about this convergence mode can be found in Rényi (1963), Aldous and Eagleson (1978), Jacod and Protter (2012) and Aït-Sahalia and Jacod (2014).

We have the following functional central limit theorem.

THEOREM 3.1 *Under assumptions 2.1 and 2.2, we have*

$$\frac{1}{\sqrt{\Delta_n}} \left(V_n(u,v) - \int_0^T e^{-\langle (u,v),((\sigma_s^X)^2,(\sigma_s^Y)^2)\rangle} \mathrm{d}s \right) \xrightarrow{\mathcal{S}} \Phi_T(u,v), \tag{4}$$

where the convergence is on the space $\mathcal{C}(\mathbb{R}_+^2)$ of continuous functions indexed by (u,v) and equipped with the local uniform topology (i.e. uniformly over compact sets of $(u,v) \in \mathbb{R}_+^2$), and the process $\Phi_T(u,v)$ is defined on an extension of the original probability space and is an $\mathcal{F}$-conditionally Gaussian process with zero-mean function and covariance function $\int_0^T F(\sqrt{u}\sigma_s^X, \sqrt{v}\sigma_s^Y, \sqrt{u'}\sigma_s^X, \sqrt{v'}\sigma_s^Y)\mathrm{d}s$ for every $(u,v,u',v') \in \mathbb{R}_+^4$ with

$$F(x,y,\bar{x},\bar{y}) = e^{-(x^2+y^2+\bar{x}^2+\bar{y}^2)} \cdot \left\{ e^{-2(x\bar{x}+y\bar{y})} + e^{2(x\bar{x}+y\bar{y})} - 2 \right\}.$$

A consistent estimator for the covariance function of $\Phi_T(u,v)$ is given by

$$\tilde{\Gamma}_n = 4\Delta_n \sum_{i=1}^{\lfloor n/2 \rfloor} \cos\left(\frac{\sqrt{2u}\Delta_{2i-1}^n X + \sqrt{2v}\Delta_{2i}^n Y}{\sqrt{\Delta_n}} \right) \cos\left(\frac{\sqrt{2u'}\Delta_{2i-1}^n X + \sqrt{2v'}\Delta_{2i}^n Y}{\sqrt{\Delta_n}} \right) - 4\Delta_n \sum_{i=1}^{\lfloor n/2 \rfloor} \cos\left(\frac{\sqrt{2(u+u')}\Delta_{2i-1}^n X + \sqrt{2(v+v')}\Delta_{2i}^n Y}{\sqrt{\Delta_n}} \right).$$

The above estimator uses the non-overlapped increments, which yields a neat asymptotic covariance function. However, the estimation efficiency can be improved by using the overlapped increments. We have the following results.

THEOREM 3.2 *Under assumptions 2.1 and 2.2, we have*

$$\frac{1}{\sqrt{\Delta_n}} \left(U_n(u,v) - \int_0^T e^{-\langle (u,v),((\sigma_s^X)^2,(\sigma_s^Y)^2)\rangle} \mathrm{d}s \right) \xrightarrow{\mathcal{S}} \Phi_T(u,v), \tag{5}$$

where the convergence is on the space $\mathcal{C}(\mathbb{R}_+^2)$ continuous functions indexed by (u,v) and equipped with the local uniform topology (i.e. uniformly over compact sets of $(u,v) \in \mathbb{R}_+^2$), and the process $\Phi_T(u,v)$ is defined on an extension of the original probability space and is an $\mathcal{F}$-conditionally Gaussian process with zero-mean function and covariance function $\int_0^T F(\sqrt{u}\sigma_s^X, \sqrt{v}\sigma_s^Y, \sqrt{u'}\sigma_s^X, \sqrt{v'}\sigma_s^Y; \rho_s)\mathrm{d}s$ for every $(u,v,u',v') \in \mathbb{R}_+^4$ with

$$F(x,y,\bar{x},\bar{y};z) = \frac{1}{2}e^{-(x^2+y^2+\bar{x}^2+\bar{y}^2)} \cdot \left\{ \frac{1+e^{4x\bar{x}+4y\bar{y}}}{e^{2x\bar{x}+2y\bar{y}}} + \frac{1+e^{4\bar{x}yz}}{e^{2\bar{x}yz}} + \frac{1+e^{4x\bar{y}z}}{e^{2x\bar{y}z}} - 6 \right\}.$$

A consistent estimator of the asymptotic covariance function is given by

$$\begin{aligned}\tilde{\Gamma}_n = {} & \Delta_n \sum_{i=1}^{n-1} \cos\left(\frac{\sqrt{2u}\Delta_i^n X + \sqrt{2v}\Delta_{i+1}^n Y}{\sqrt{\Delta_n}} \right) \cos\left(\frac{\sqrt{2u'}\Delta_i^n X + \sqrt{2v'}\Delta_{i+1}^n Y}{\sqrt{\Delta_n}} \right) \\ & + \Delta_n \sum_{i=1}^{n-2} \cos\left(\frac{\sqrt{2u}\Delta_{i-1}^n X}{\sqrt{\Delta_n}} \right) \cos\left(\frac{\sqrt{2v}\Delta_{i+1}^n Y}{\sqrt{\Delta_n}} \right) \cos\left(\frac{\sqrt{2u'}\Delta_{i+1}^n X}{\sqrt{\Delta_n}} \right) \cos\left(\frac{\sqrt{2v'}\Delta_{i+2}^n Y}{\sqrt{\Delta_n}} \right) \\ & + \Delta_n \sum_{i=1}^{n-2} \cos\left(\frac{\sqrt{2u'}\Delta_{i-1}^n X}{\sqrt{\Delta_n}} \right) \cos\left(\frac{\sqrt{2v'}\Delta_{i+1}^n Y}{\sqrt{\Delta_n}} \right) \cos\left(\frac{\sqrt{2u}\Delta_{i+1}^n X}{\sqrt{\Delta_n}} \right) \cos\left(\frac{\sqrt{2v}\Delta_{i+2}^n Y}{\sqrt{\Delta_n}} \right) \\ & - 3\Delta_n \sum_{i=1}^{n-1} \cos\left(\frac{\sqrt{2(u+u')}\Delta_i^n X + \sqrt{2(v+v')}\Delta_{i+1}^n Y}{\sqrt{\Delta_n}} \right).\end{aligned}$$

REMARK 1 We can compare the asymptotic efficiency of two estimators explicitly. $U_n(u,v)$ makes good use of the data compared to $V_n(u,v)$ and should have smaller asymptotic variance. We want to prove

$$\begin{aligned} & e^{-2(u(\sigma_s^X)^2+v(\sigma_s^Y)^2)} \cdot \left\{ e^{-2(u(\sigma_s^X)^2+v(\sigma_s^Y)^2)} + e^{2(u(\sigma_s^X)^2+v(\sigma_s^Y)^2)} - 2 \right\} \\ & > \frac{1}{2}e^{-2(u(\sigma_s^X)^2+v(\sigma_s^Y)^2)} \cdot \\ & \times \left\{ e^{-2(u(\sigma_s^X)^2+v(\sigma_s^Y)^2)} + e^{2(u(\sigma_s^X)^2+v(\sigma_s^Y)^2)} + 2\left(e^{-2\sqrt{uv}\sigma_s^X\sigma_s^Y\rho_s} + e^{2\sqrt{uv}\sigma_s^X\sigma_s^Y\rho_s} \right) - 6 \right\}. \end{aligned}$$

For convenience of expression, we let,

$$a := e^{-2(u(\sigma_s^X)^2+v(\sigma_s^Y)^2)} + e^{2(u(\sigma_s^X)^2+v(\sigma_s^Y)^2)} - 2,$$

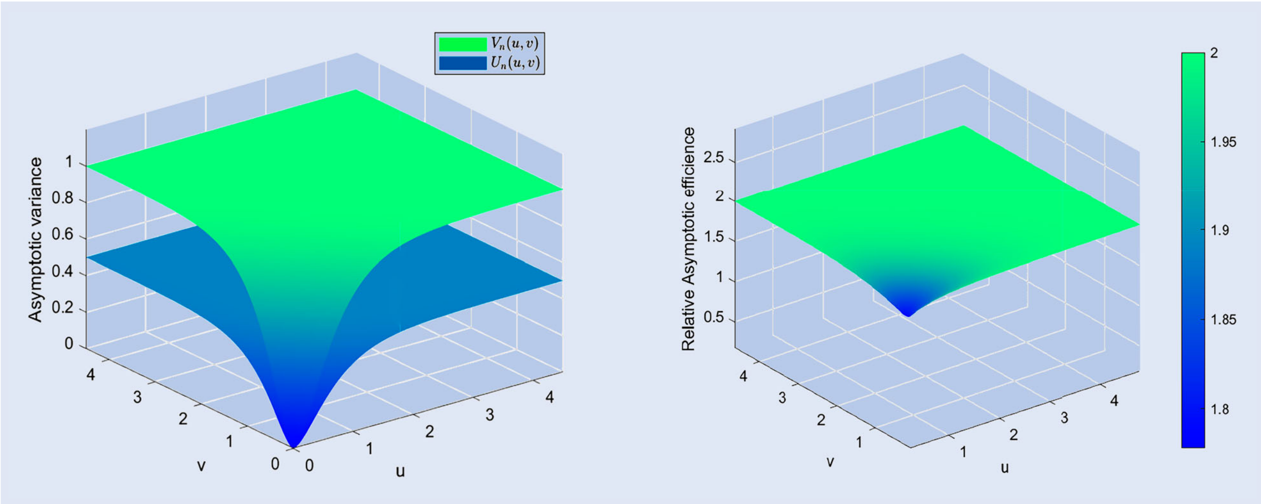

Figure 1. The asymptotic variance of $V_n(u, v)$ and $U_n(u, v)$.

$$b := 2\left(e^{-2\sqrt{uv}\sigma_s^X\sigma_s^Y\rho_s} + e^{2\sqrt{uv}\sigma_s^X\sigma_s^Y\rho_s}\right) - 4.$$

It suffices to prove $a > b$. We define $c := u(\sigma_s^X)^2 + v(\sigma_s^Y)^2$, $d := 2\sqrt{uv}\sigma_s^X\sigma_s^Y$, $c, d \in \mathbb{R}^+$. Hence $c \geq d$, $|d\rho_s| \leq |d|$, and

$$\begin{aligned} a &= e^{-2c} + e^{2c} - 2 \\ &\geq e^{-2d} + e^{2d} - 2 \\ &> 2(e^{-d} + e^{d}) - 4 \\ &\geq 2(e^{-d\rho_s} + e^{d\rho_s}) - 4 = b. \end{aligned}$$

We can also visualize two asymptotic variances. In figure 1, we display the plots of two asymptotic variances as a function of (u, v). In the figure, we let $T = 1, \sigma_s^X = 1, \sigma_s^Y = 1, u = u'$, $v = v'$ and $\rho = 0.5$. From figure 1, we see that the asymptotic variance of $U_n(u, v)$ is uniformly smaller than that of $V_n(u, v)$.

Although $U_n(u, v)$ is better than $V_n(u, v)$ in the sense of estimation performance, we still keep $V_n(u, v)$ in the paper for the following reasons. First, it is seen that the asymptotic variance of $V_n(u, v)$ does not depend on the correlation process ρ_t of X and Y. Thus it is robust to the dependence between two latent processes. Second, we can get an improved version of $V_n(u, v)$ by adding the other symmetric part, namely,

$$V_n'(u,v) = \Delta_n \sum_{i=1}^{\lfloor n/2 \rfloor}\left[\cos\left(\frac{\sqrt{2u}\Delta_{2i-1}^n X + \sqrt{2v}\Delta_{2i}^n Y}{\sqrt{\Delta_n}}\right) + \cos\left(\frac{\sqrt{2v}\Delta_{2i-1}^n Y + \sqrt{2u}\Delta_{2i}^n X}{\sqrt{\Delta_n}}\right)\right].$$

We can get the central limit theory of $V_n'(u, v)$ as the following:

$$\frac{1}{\sqrt{\Delta_n}}\left(V_n'(u,v) - \int_0^T \mathrm{e}^{-\langle (u,v),((\sigma_s^X)^2,(\sigma_s^Y)^2)\rangle}\mathrm{d}s\right) \xrightarrow{\mathcal{S}} \Phi_T(u,v), \tag{6}$$

where the process $\Phi_T(u, v)$ is the same as the limit process of $U_n(u, v)$. We also include this estimator in the simulation study in section 5. However, using a similar modification to the estimator $U_n(u, v)$ does not improve the estimation efficiency.

4. Joint infill and long-time span asymptotics

In this section, we extend the above analysis (for fixed T) and consider the asymptotic results when $T \to \infty$ and $\Delta_n \to 0$. To facilitate the derivation of this theory, we require some additional notations. We define

$$\begin{aligned} Z_t^{x,y}(u,v) &= \int_{t-1}^t e^{-u(\sigma_s^x)^2 - v(\sigma_s^y)^2}\mathrm{d}s, \quad \hat{Z}_t^{x,y}(u,v) \\ &= \sum_{\lfloor (t-1)/\Delta_n \rfloor + 1}^{\lfloor t/\Delta_n \rfloor} \Delta_n \cos\left(\frac{\sqrt{2u}\Delta_i^n X + \sqrt{2v}\Delta_{i+1}^n Y}{\sqrt{\Delta_n}}\right), \\ Z_t^x(u) &= \int_{t-1}^t e^{-u(\sigma_s^x)^2}\mathrm{d}s, \quad \hat{Z}_t^x(u) \\ &= \sum_{\lfloor (t-1)/\Delta_n \rfloor + 1}^{\lfloor t/\Delta_n \rfloor} \Delta_n \cos\left(\frac{\sqrt{2u}\Delta_i^n X}{\sqrt{\Delta_n}}\right), \\ Z_t^y(v) &= \int_{t-1}^t e^{-u(\sigma_s^y)^2}\mathrm{d}s, \quad \hat{Z}_t^y(v) \\ &= \sum_{\lfloor (t-1)/\Delta_n \rfloor + 1}^{\lfloor t/\Delta_n \rfloor} \Delta_n \cos\left(\frac{\sqrt{2u}\Delta_i^n Y}{\sqrt{\Delta_n}}\right), \end{aligned}$$

and $\mu_t^{x,y}(u, v) := \mathbb{E}[Z_t^{x,y}(u, v)]$, $\mu_t^x(u) := \mathbb{E}[Z_t^x(u)]$, $\mu_t^y(v) := \mathbb{E}[Z_t^y(v)]$. We need the following assumption to derive the long-time span asymptotic behavior.

Assumption 4.1 The volatility processes σ_t^Z, $Z = X, Y$ are stationary and α-mixing processes with coefficient $\alpha^{mix} = O(t^{-\gamma})$ for some $\gamma > 1$ when $t \to \infty$, where

$$\begin{aligned} \alpha_t^{mix} &= \sup_{A\in\mathcal{F}_0, B\in\mathcal{F}^t} |\mathbb{P}(A\cap B) - \mathbb{P}(A)\mathbb{P}(B)|, \\ \mathcal{F}_0 &= \sigma(\sigma_s^Z, W_s^Z, Z = X, Y, s \leq 0) \quad \text{and} \\ \mathcal{F}^t &= \sigma(\sigma_s^Z, W_s^Z - W_t^Z, Z = X, Y, s \geq t). \end{aligned}$$

Due to the assumption of stationarity, we hence have

$$\mu_t^{x,y}(u,v) = \mu_1^{x,y}(u,v), \quad \mu_t^x(u) = \mu_1^x(u), \quad \mu_t^y(v) = \mu_1^y(v).$$

To control the approximation error by ignoring the drift term and jump part in the long span, we further require the following assumption of uniform boundedness, which is same as assumption 1 of Christensen *et al.* (2019).

ASSUMPTION 4.2 We assume that for each $p \geq 1$ and $t \geq 0$, $\mathbb{E}[|b_t^Z|^p] + \mathbb{E}[|\sigma_t^Z|^p] + \int_R (1 \vee \delta^Z)^p \mu^Z(dt, dZ) \leq C$, where $Z = X, Y$. In addition, σ_t^Z is an Itô semi-martingale given by

$$\sigma_t^Z = \sigma_0^Z + \int_0^t \tilde{b}_s^Z \mathrm{d}s + \int_0^t v_s^Z \mathrm{d}W_s^Z + \int_0^t v_s'^Z \mathrm{d}W_s'^Z + \int_0^t \int_{\mathbb{R}} \delta'^Z(s-, z)\mu'^Z(\mathrm{d}s, \mathrm{d}z),$$

where $\mathbb{E}[|\tilde{b}_s^Z|^p] + \mathbb{E}[|v_s^Z|^p] + \mathbb{E}[|v_s'^Z|^p] \leq C$ for some constant C. $\delta'^Z(s-, z)$ is a predictable function, and $\delta'^Z(\omega, s-, z) \wedge 1 \leq \tilde{\Gamma}(z)$ for all (ω, t, z) and some $\tilde{\Gamma} : \mathbb{R} \to \mathbb{R}$ such that $\int_{\mathbb{R}} \tilde{\Gamma}(z)^2 \lambda(dz) < \infty$

The stationary and mixing condition in assumption 4.1 is required to prove the central limit theorem under the long-span setting. In short, we hope to restrict the memory of the volatilities so that the volatility processes are not 'too' strongly dependent. This still can capture a wide variety of volatility models, such as a large class of processes driven by Brownian motion (e.g. Heston 1993) or the Lévy driven Ornstein–Uhlenbeck model of Barndorff-Nielsen and Shephard (2001), where volatility is governed by general (positive) processes.

THEOREM 4.3 *Under assumptions 2.1–4.1, if $T \to \infty$, $\Delta_n \to 0$ and $T\Delta_n \to 0$, we have*

$$\sqrt{T} \begin{pmatrix} \frac{1}{T}\sum_{t=1}^{T} \hat{Z}_t^{x,y}(u,v) - \mu_1^{x,y}(u,v) \\ \frac{1}{T}\sum_{t=1}^{T} \hat{Z}_t^{x}(u) - \mu_1^{x}(u) \\ \frac{1}{T}\sum_{t=1}^{T} \hat{Z}_t^{y}(v) - \mu_1^{y}(v) \end{pmatrix} \xrightarrow{\mathcal{W}} \Phi(u,v),$$

where the convergence is on the space $\mathcal{C}(\mathbb{R}_+^2)$ of continuous functions indexed by (u,v) and equipped with the local uniform topology (i.e. uniformly over compact sets of $(u,v) \in \mathbb{R}_+^2$), and the process $\Phi(u,v)$ is defined on an extension of the original probability space and is an $\mathcal{F}$-conditionally Gaussian process with zero-mean function and covariance function matrices $V([u,v],[u',v'])$ for every $(u,v,u',v') \in \mathbb{R}_+^4$ with

$$V_{ab}([u,v],[u',v']) = \mathbb{E}[A_a(0)A_b'(0)] + 2\sum_{l=1}^{\infty}\mathbb{E}[A_a(0)A_b'(l)], \tag{7}$$

here $a, b = 1, 2, 3, a \leq b$ and,

$$A_1(l) = Z_{l+1}^{x,y}(u,v) - \mu_1^{x,y}(u,v),$$
$$A_1'(l) = Z_{l+1}^{x,y}(u',v') - \mu_1^{x,y}(u',v'),$$
$$A_2(l) = Z_{l+1}^{x}(u) - \mu_1^{x}(u), \quad A_2'(l) = Z_{l+1}^{x}(u') - \mu_1^{x}(u'),$$
$$A_3(l) = Z_{l+1}^{y}(v) - \mu_1^{y}(u), \quad A_3'(l) = Z_{l+1}^{y}(v') - \mu_1^{y}(v').$$

Moreover, if L_T is a deterministic sequence of integers satisfying $\frac{L_T}{\sqrt{T}} \to 0$ and $L_T\Delta_n^{1/2} \to 0$ as $T \to \infty$, $\Delta_n \to 0$, a consistent estimator of the asymptotic covariance function matrices is given by $\hat{V}([u,v],[u',v'])$ for every $(u,v,u',v') \in \mathbb{R}_+^4$ with

$$\hat{V}_{ab}([u,v],[u',v']) = \hat{C}_{ab}^0([u,v],[u',v']) + \sum_{i=1}^{L_T}\omega(i, L_T)(\hat{C}_{ab}^i([u,v],[u',v']) + \hat{C}_{ab}^i([u',v'],[u,v])), \tag{8}$$

where $a, b = 1, 2, 3, a \leq b$. The $\omega(i, L_T)$ is either a Bartlett or Parzen Kernel. In addition,

$$\hat{C}_{ab}^l([u,v],[u',v']) = \frac{1}{T}\sum_{t=l+1}^{T} B_{at}(0)B_{bt}'(l),$$

with

$$B_{1t}(l) = \hat{Z}_{t-l}^{x,y}(u,v) - \frac{1}{T}\sum_{t=1}^{T}\hat{Z}^{x,y}(u,v),$$

$$B_{1t}'(l) = \hat{Z}_{t-l}^{x,y}(u',v') - \frac{1}{T}\sum_{t=1}^{T}\hat{Z}^{x,y}(u',v'),$$

$$B_{2t}(l) = \hat{Z}_{t-l}^{x}(u) - \frac{1}{T}\sum_{t=1}^{T}\hat{Z}^{x}(u),$$

$$B_{2t}'(l) = \hat{Z}_{t-l}^{x}(u') - \frac{1}{T}\sum_{t=1}^{T}\hat{Z}^{x}(u'),$$

$$B_{3t}(l) = \hat{Z}_{t-l}^{y}(v) - \frac{1}{T}\sum_{t=1}^{T}\hat{Z}^{y}(v),$$

$$B_{3t}'(l) = \hat{Z}_{t-l}^{y}(v') - \frac{1}{T}\sum_{t=1}^{T}\hat{Z}^{y}(v').$$

Finally, an application of the Delta method yields the following result.

COROLLARY 4.4 *Under the same conditions and notations as theorem 4.3, we have*

$$\sqrt{T}\left(\frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^{x,y}(u,v) - \frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^{x}(u)\frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^{y}(v) - \left(\mu_1^{x,y}(u,v) - \mu_1^{x}(u)\mu_1^{y}(v)\right)\right) \xrightarrow{\mathcal{W}} \boldsymbol{\gamma} \cdot \Phi(u,v),$$

where $\boldsymbol{\gamma} = [1, -\mu_1^{y}(v), -\mu_1^{x}(u)]$.

Now we use the above result to construct a test for the dependence of volatilities. That is, we want to test

$$H_0 : \sigma_s^x \text{ and } \sigma_s^y \text{ are independent} \quad \text{vs.}$$
$$H_1 : \sigma_s^x \text{ and } \sigma_s^y \text{ are dependent.}$$

Under H_0, for all $(u, v) \in \mathbb{R}_+^2$, we have

$$\mu_1^{x,y}(u, v) - \mu_1^x(u)\mu_1^y(v) = 0,$$

hence by corollary 4.4, we have

$$\hat{\mathcal{S}}_{T,n} := \sqrt{T}\left(\frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^{x,y}(u, v) - \frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^x(u)\frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^y(v)\right) \xrightarrow{\mathcal{W}} \boldsymbol{\gamma}\cdot\Phi(u, v).$$

According to the Continuous Mapping Theorem,

$$||\hat{\mathcal{S}}_{T,n}||^2 = \iint_{\mathbb{R}_+^2}[\hat{\mathcal{S}}_{T,n}(u, v)]^2\mathrm{d}u\mathrm{d}v \xrightarrow{d} \iint[\boldsymbol{\gamma}\cdot\boldsymbol{\Phi}(u, v)]^2\mathrm{d}u\mathrm{d}v,$$

where the norm $||\cdot||$ is defined in the Hilbert space $\mathcal{L}^2$,

$$\mathcal{L}^2 = \left\{f : \mathbb{R}_+^2 \to \mathbb{R}\left|\iint_{\mathbb{R}_+^2} f(u, v)^2\mathrm{d}u\mathrm{d}v < \infty\right.\right\}.$$

Therefore, we use $||\hat{\mathcal{S}}_{T,n}||^2$ as the test statistic and observe that under H_1, $||\hat{\mathcal{S}}_{T,n}||^2 \to \infty$, hence the critical region is

$$C = \{||\hat{\mathcal{S}}_{T,n}||^2 > d_\alpha\},$$

where d_α can be determined by the distribution of $||\boldsymbol{\gamma}\cdot\boldsymbol{\Phi}(u, v)||^2$. To approximate the distribution of $||\boldsymbol{\gamma}\cdot\boldsymbol{\Phi}(u, v)||^2$, we first partition the rectangle $[0, 1]\times[0, 1]$ into $m\times m$ (m is a large positive integer) same sub-rectangle, and let $u_i = i/m, v_j = j/m, i, j = 1, 2\dots m$. Then we approximate the limit distribution of the test statistic by

$$\int_0^1\int_0^1[\boldsymbol{\gamma}(u, v)\cdot\boldsymbol{\Phi}(u, v)]^2\,\mathrm{d}u\mathrm{d}v \approx \frac{1}{m^2}\sum_{i=1}^{m}\sum_{j=1}^{m}\left[\boldsymbol{\gamma}(u_i, v_j)\cdot\boldsymbol{\Phi}(u_i, v_j)\right]^2, \tag{9}$$

where $(\boldsymbol{\gamma}(u_i, v_j)\cdot\boldsymbol{\Phi}(u_i, v_j))_{i,j}$ is conditionally on $\mathcal{F}$, an $m\times m$ dimensional normal random variable with conditional covariance defined in (7). Hence we can estimate the covariance between $(\boldsymbol{\gamma}(u_i, v_j)\cdot\boldsymbol{\Phi}(u_i, v_j))$ and $(\boldsymbol{\gamma}(u_{i'}, v_{j'})\cdot\boldsymbol{\Phi}(u_{i'}, v_{j'}))_{i',j'}$ using the estimator given by (8), denoted as $\hat{C}((u_i, v_j), (u_{i'}, v_{j'}))$. Hence, if we compute the eigenvalues of $(\hat{C}((u_i, v_j), (u_{i'}, v_{j'})))_{ij,i'j'}$, denoted as $\hat{\pi}_k, k = 1,\dots, m^2$, then the distribution of $\int_0^1\int_0^1[\boldsymbol{\gamma}(u, v)\cdot\boldsymbol{\Phi}(u, v)]^2\mathrm{d}u\mathrm{d}v$ can be approximated by

$$\mathcal{M} := \frac{1}{m^2}\sum_{i=1}^{m^2}\hat{\pi}_k\chi_k^2,$$

where χ_k^2's are independent Chi-squared random variables of degree of freedom one. The distribution of $\mathcal{M}$ can be simulated by the Monte Carlo method.

PROPOSITION 4.5 *The test statistic* $||\hat{\mathcal{S}}_{T,n}||^2$ *has asymptotic size* α *under the null hypothesis and is consistent under the fixed alternative hypothesis* H_1*, that is*

$$\mathbb{P}(||\hat{\mathcal{S}}_{T,n}||^2 > \hat{d}_\alpha|H_0) \to \alpha, \quad \mathbb{P}(||\hat{\mathcal{S}}_{T,n}||^2 > \hat{d}_\alpha|H_1) \to 1,$$

where $\hat{d}_\alpha$ *is the upper* α*-quantile of the distribution of* $\mathcal{M}$.

5. Simulation studies

In this section, we conduct some simulation studies to examine the finite sample performance of our proposed estimators and the test.

5.1. Finite sample performance of U_n and V_n

Throughout this simulation, we consider the fixed time interval $[0, T]$ with $T = 1$ and observe the process at some discrete time points $\{i\Delta_n, i = 1, 2,\dots, n\}$, $n = 1760$, which corresponds to monthly observations with 5-minute frequency. We study the following three commonly used stochastic volatility models.

EXAMPLE 5.1 We generate the high-frequency data $\{(X_t, Y_t), t \in [0, T]\}$ from the following processes:

$$\begin{cases}\mathrm{d}X_t = 0.03\mathrm{d}t + \sigma_t^X\mathrm{d}W_t^X,\\ \mathrm{d}Y_t = 0.04\mathrm{d}t + \sigma_t^Y\mathrm{d}W_t^Y,\end{cases}$$

where $\mathrm{d}W_t^Y = \rho_t\mathrm{d}W_t^X + \sqrt{1-\rho_t^2}\mathrm{d}W_t^\star$, W_t^X and $W_t^\star$ are two mutually independent Brownian motions. Without loss of generality, we let $\rho_t = 0.5$ throughout the simulation study. Moreover, $\sigma_t^X = \exp\{0.3125 - 0.125\tau_t^X\}$, $\mathrm{d}\tau_t^X = -0.025\tau_t^X\mathrm{d}t + \mathrm{d}B_t^X$, and $\sigma_t^Y = \exp\{0.4500 - 0.325\tau_t^Y\}$, $\mathrm{d}\tau_t^Y = -0.030\tau_t^Y\mathrm{d}t + \mathrm{d}B_t^Y$. Here, W_t^Z and B_t^Z, $Z = X, Y$ are independent of each other.

EXAMPLE 5.2 $\{(X_t, Y_t), t \in [0, T]\}$ follows a stochastic volatility model with Poisson jumps:

$$\begin{cases}\mathrm{d}X_t = 0.03\mathrm{d}t + \sigma_t^X\mathrm{d}W_t^X + \sum_{i=1}^{N_t^X} Y_i,\\ \mathrm{d}Y_t = 0.04\mathrm{d}t + \sigma_t^Y\mathrm{d}W_t^Y + \sum_{i=1}^{N_t^Y} Y_i,\end{cases}$$

where N_t^X and N_t^Y are Poisson processes with $\mathbb{E}[N_t^X] = 2t$, $\mathbb{E}[N_t^Y] = 3t$. Furthermore, we set the jump sizes $Y_i \overset{i.i.d}{\sim} N(0, 1)$. The generation of σ_t^X, σ_t^Y, $\mathrm{d}W_t^X$ and $\mathrm{d}W_t^Y$ are the same as in Example 5.1.

EXAMPLE 5.3 $\{(X_t, Y_t), t \in [0, T]\}$ follows a stochastic volatility model with α-stable jumps:

$$\begin{cases}\mathrm{d}X_t = 0.03\mathrm{d}t + \sigma_t^X\mathrm{d}W_t^X + \mathrm{d}J_t^X,\\ \mathrm{d}Y_t = 0.04\mathrm{d}t + \sigma_t^Y\mathrm{d}W_t^Y + \mathrm{d}J_t^Y,\end{cases}$$

Table 1. Monte Carlo results of the estimators $V_n(u,v)$, $U_n(u,v)$ and $V_n'(u,v)$.

		$u=2.5$			$u=3.5$			$u=4.5$		
$V_n(u,v)$		Bias	SD	MSE	Bias	SD	MSE	Bias	SD	MSE
					Example 1					
v	2.75	0.00024	0.02419	0.00059	0.00077	0.02390	0.00057	− 0.00050	0.02465	0.00061
	3.75	0.00046	0.02415	0.00058	− 0.00014	0.02398	0.00058	0.00116	0.02447	0.00060
	4.75	0.00083	0.02384	0.00057	− 0.00113	0.02352	0.00055	− 0.00141	0.02430	0.00059
					Example 2					
v	2.75	− 0.00023	0.02361	0.00056	0.00184	0.02456	0.00061	− 0.00067	0.02355	0.00056
	3.75	0.00048	0.02298	0.00053	0.00112	0.02271	0.00052	0.00012	0.02329	0.00054
	4.75	− 0.00012	0.02386	0.00057	0.00053	0.02393	0.00057	− 0.00126	0.02385	0.00057
					Example 3					
v	2.75	− 0.00036	0.02337	0.00055	− 0.00081	0.02384	0.00057	0.00047	0.02255	0.00051
	3.75	− 0.00071	0.02344	0.00055	0.00082	0.02349	0.00055	− 0.00056	0.02500	0.00063
	4.75	− 0.00027	0.02412	0.00058	− 0.00011	0.02371	0.00056	− 0.00048	0.02473	0.00061

		$u=2.5$			$u=3.5$			$u=4.5$		
$U_n(u,v)$		Bias	SD	MSE	Bias	SD	MSE	Bias	SD	MSE
					Example 1					
v	2.75	0.00071	0.01670	0.00028	0.00040	0.01713	0.00029	− 0.00031	0.01667	0.00028
	3.75	− 0.00042	0.01655	0.00027	0.00042	0.01655	0.00027	− 0.00052	0.01713	0.00029
	4.75	− 0.00005	0.01644	0.00027	− 0.00085	0.01608	0.00026	− 0.00011	0.01713	0.00029
					Example 2					
v	2.75	− 0.00015	0.01670	0.00028	0.00049	0.01682	0.00028	0.00020	0.01638	0.00027
	3.75	− 0.00022	0.01599	0.00026	− 0.00019	0.01705	0.00029	0.00007	0.01637	0.00027
	4.75	− 0.00063	0.01695	0.00029	− 0.00058	0.01660	0.00028	0.00012	0.01671	0.00028
					Example 3					
v	2.75	− 0.00013	0.01753	0.00031	0.00011	0.01647	0.00027	− 0.00027	0.01679	0.00028
	3.75	0.00032	0.01679	0.00028	− 0.0001	0.01726	0.00030	0.00064	0.01746	0.00031
	4.75	− 0.00018	0.01648	0.00027	0.00048	0.01621	0.00026	− 0.00067	0.01734	0.00030

		$u=2.5$			$u=3.5$			$u=4.5$		
$V_n'(u,v)$		Bias	SD	MSE	Bias	SD	MSE	Bias	SD	MSE
					Example 1					
v	2.75	− 0.00038	0.01670	0.00028	− 0.00028	0.01717	0.00029	− 0.00076	0.01697	0.00029
	3.75	0.00057	0.01654	0.00027	0.00064	0.01671	0.00028	0.00054	0.01644	0.00027
	4.75	− 0.00008	0.01689	0.00029	− 0.00005	0.01673	0.00028	− 0.00081	0.01654	0.00027
					Example 2					
v	2.75	− 0.00020	0.01660	0.00028	− 0.00023	0.01679	0.00028	0.00022	0.01696	0.00029
	3.75	0.00022	0.01643	0.00027	0.00029	0.01697	0.00029	0.00015	0.01676	0.00028
	4.75	− 0.00071	0.01683	0.00028	− 0.00006	0.01708	0.00029	0.00062	0.01660	0.00028
					Example 3					
v	2.75	− 0.00053	0.01719	0.00030	− 0.00031	0.01709	0.00029	0.00024	0.01626	0.00026
	3.75	− 0.00049	0.01684	0.00028	− 0.00043	0.01653	0.00027	− 0.00032	0.01647	0.00027
	4.75	0.00015	0.01678	0.00028	0.00030	0.01691	0.00029	− 0.00048	0.01669	0.00028

where σ_t^X, σ_t^Y, dW_t^X and dW_t^Y are the same as in Example 5.1, and J_t^X, J_t^Y are both the symmetric α-stable processes with $\alpha_1 = 0.5$ and $\alpha_2 = 0.9$ respectively.

The simulations are repeated 5000 times. We compute the bias, standard deviation (SD), and mean square error (MSE) of our proposed estimators, $V_n(u,v)$, $U_n(u,v)$, and $V_n'(u,v)$. Results are displayed in table 1.

We assess the asymptotic normality of the estimators $V_n(u,v)$ and $U_n(u,v)$ via both the infeasible and feasible limit theorems, respectively, and set $u = 3.5$, $v = 3.75$. In figure 2, we display the histograms of the infeasible statistics, and the histograms of the studentized statistic are exhibited in figure 3, for the setting of Example 5.3, the results are similar for other settings.

We have the following observations from this simulation.

(i) The bias of estimators $V_n(u,v)$, $U_n(u,v)$, $V_n'(u,v)$ are very close to zero, which demonstrates all of our estimators are consistent.

(ii) From table 1, $U_n(u,v)$ performs better than $V_n(u,v)$ in terms of overall mean square error. Compared to $V_n(u,v)$, $U_n(u,v)$ always has smaller SD and Bias. This is in line with our theory.

(iii) We see that the results for $U_n(u,v)$ and $V_n'(u,v)$ are very similar from table 1, which is consistent with the limiting theory we obtained in section 3.

(iv) All estimators are robust to the Poisson jumps and α-stable jumps with $\alpha < 1$.

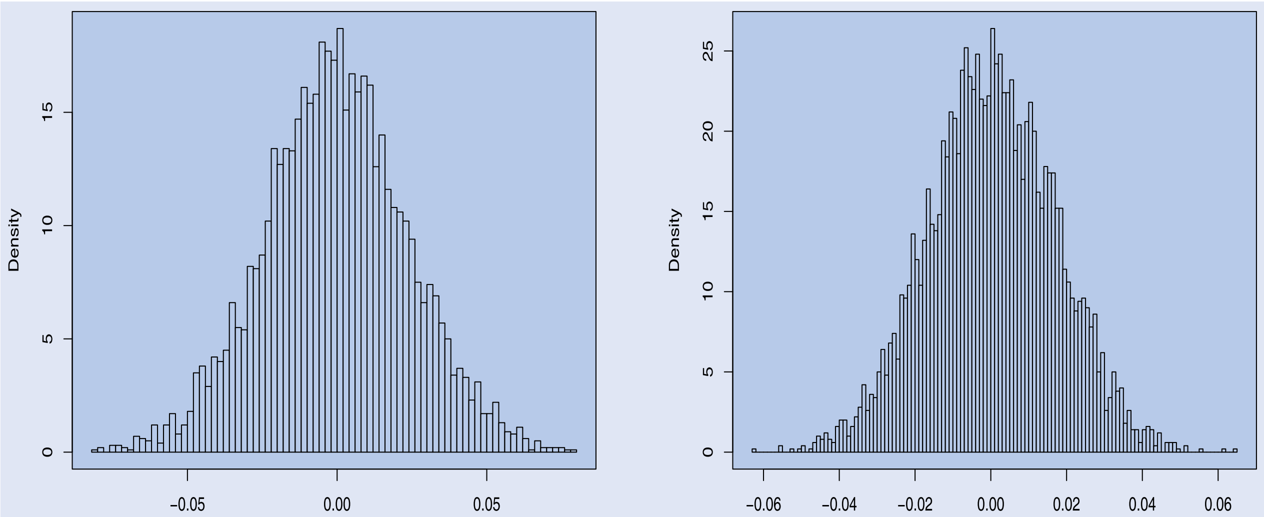

Figure 2. Left panel: The histogram of $V_n(u,v) - \int_0^1 \mathrm{e}^{-\langle (u,v),((\sigma_s^X)^2,(\sigma_s^Y)^2)\rangle}\mathrm{d}s$; Right panel: The histogram of $U_n(u,v) - \int_0^1 \mathrm{e}^{-\langle (u,v),((\sigma_s^X)^2,(\sigma_s^Y)^2)\rangle}\mathrm{d}s$.

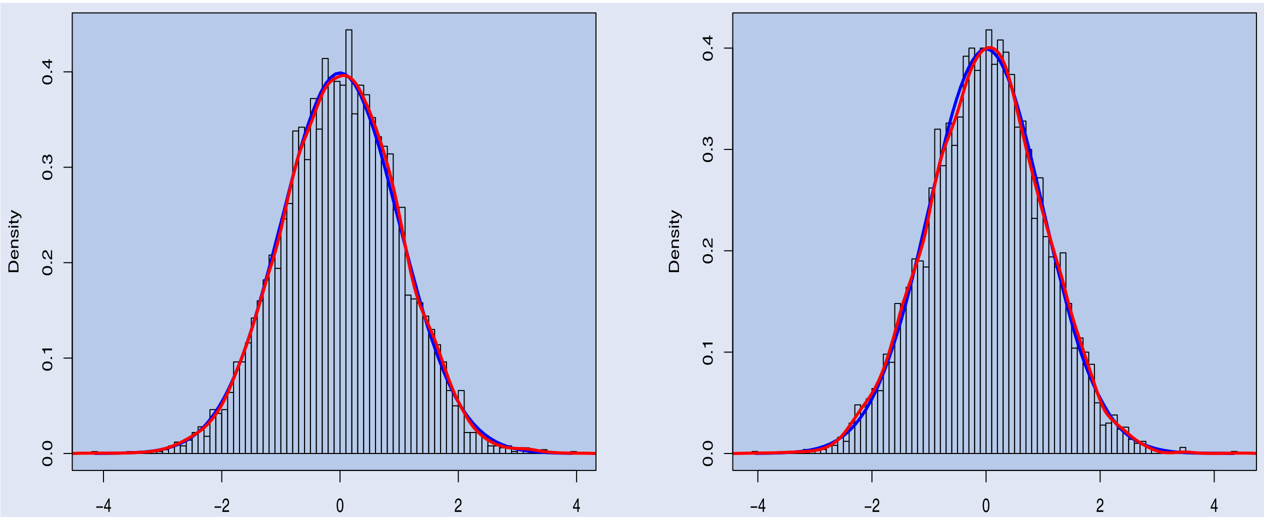

Figure 3. Left panel: The histogram of $\frac{1}{\sqrt{\Delta_n \tilde{\Gamma}_n}}(V_n(u,v) - \int_0^1 \mathrm{e}^{-\langle (u,v),((\sigma_s^X)^2,(\sigma_s^Y)^2)\rangle}\mathrm{d}s)$; Right panel: The histogram of $\frac{1}{\sqrt{\Delta_n \tilde{\Gamma}_n}}(U_n(u,v) - \int_0^1 \mathrm{e}^{-\langle (u,v),((\sigma_s^X)^2,(\sigma_s^Y)^2)\rangle}\mathrm{d}s)$.

(v) Figures 2 and 3 show that the estimators behave like a normal distribution, justifying our asymptotic theory.

5.2. *Test for the dependence of volatility processes*

In this simulation study, we use the test statistic $||\hat{S}_{T,n}||^2$, to test the dependence of volatilities. We consider the following data-generating process.

Example 5.4 We use the data-generating processes of Example 5.1, with τ_t^X and τ_t^Y generated from the following discrete auto-regressive processes. That is, we consider

$$\tau_t^x = 0.5\tau_{t-1}^x + \epsilon_t^x, \quad \tau_t^y = 0.7\tau_{t-1}^y + \epsilon_t^y, \quad \text{with}$$
$$\epsilon_t^y = \rho_t' \epsilon_t^x + \sqrt{1-\rho_t'^2}\epsilon_t^\star,$$

where ϵ_t^x and $\epsilon_t^\star$ are two mutually independent standard normal random variables. It follows that ϵ_t^y is independent of ϵ_t^x if $\rho_t' = 0$, otherwise they are dependent.

Hence, we want to test the following hypothesis:

$$H_0 : \rho_t' \equiv 0, \quad \text{vs.} \quad H_1 : \rho_t' \not\equiv 0.$$

Under the null hypothesis, we set $m = 10$ to simulate the limiting distribution. Specifically, we first calculate the $m^2 \times m^2$-dim conditional covariance matrix

$$\begin{aligned}
&\hat{C}((u_i, v_j), (u_{i'}, v_{j'})) \\
&= [1, -\hat{Z}_1^y(v_j), -\hat{Z}_1^x(u_i)] \\
&\quad \cdot \hat{V}([u_i, v_j], [u_{i'}, v_{j'}]) \cdot [1, -\hat{Z}_1^y(v_{j'}), -\hat{Z}_1^x(u_{i'})]^\top,
\end{aligned}$$

Table 2. The performance of test statistic. In the case TV1, we let $\rho_t' = \frac{3}{5}|\cos(t)| + 0.1$, and in TV2, we let $\rho_t' = -\frac{4}{5}|\sin(t)| - 0.1$.

	$T, n, \alpha = 0.05$				$T, n, \alpha = 0.10$			
ρ'	22, 390	44, 390	44, 780	66, 780	22, 390	44, 390	44, 780	66, 780
0	0.06	0.04	0.04	0.06	0.10	0.12	0.11	0.08
0.2	0.26	0.22	0.45	0.72	0.32	0.36	0.66	0.80
0.5	0.47	0.52	0.91	0.96	0.63	0.74	0.96	0.99
−0.5	0.51	0.59	0.89	0.99	0.63	0.74	0.94	0.99
0.8	0.65	0.82	0.99	1.00	0.75	0.92	1.00	1.00
TV1	0.27	0.30	0.63	0.78	0.43	0.51	0.71	0.87
TV2	0.42	0.57	0.85	0.97	0.52	0.71	0.89	0.98

where $1 \le i, j, i', j' \le 10$, and then compute the eigenvalues of $\hat{C}$, denoted as $\hat{\pi}_k$, $k = 1, \ldots, m^2$. Then, we generate m^2-dimensional independent χ^2 random variables with degree of freedom 1 and let

$$\mathcal{M}_1 = \frac{1}{m^2} \sum_{k=1}^{m^2} \hat{\pi}_k \chi^2_{1,k}.$$

Finally, we repeat the procedure 10 000 times to obtain $\mathcal{M}_j, j = 1 \ldots, 10\,000$ and the critical value of the test is obtained from the upper αth sample quantiles of $\mathcal{M}$.† We consider the following four scenarios of the T and Δ_n,

- $T = 22$(one month), $\Delta_n = 1/390$,
- $T = 44$(two months), $\Delta_n = 1/390$,
- $T = 44$(two months), $\Delta_n = 1/780$,
- $T = 66$(one quarter), $\Delta_n = 1/780$.

And we consider the different scenarios for the alternative hypothesis, including the constant and time-varying ρ_t'. The size and power of the test are exhibited in table 2.

From table 2, we see that the results shows that the performance of the test is satisfactory and the simulation studies are in line with our asymptotic results.

6. Empirical study

In this section, we apply our proposed test to a high-frequency data set consisting of 20 stocks of the S&P 500 index (INX). The period of the data set used is from October 1 to 31, 2019, and the sampling frequency is one minute. We selected five stocks from each of the following four sectors:

- Information technology,
- Finance,
- Consumer staples,
- Real estate.

The symbol of chosen stocks is shown in table 3. For the convenience of calculation, we rescale the returns by multiplying by 15 so that the resulting spot volatility is not affected by the approximation of the cosine function while the correlation between volatilities remains unchanged. We utilize the test statistic introduced in section 4 and obtain the critical values following the procedure in section 5. The results are presented in table 3.

From table 3, we see that at the nominal level $\alpha = 0.05$, the null hypotheses of about 75% pairs have been rejected. Namely, about 3 quarters of the stocks are dependent on each other among these 20 stocks. Furthermore, the dependence patterns within sectors and between different sectors are also different. For example, the volatilities of stocks in the sector Consumer Staples are more dependent on other sectors, showing the smallest averaged p-values. This phenomenon is probably due to the high market capitalizations of these companies and their significant connections to the daily lives of individuals. The volatilities of stocks in the Real Estate sector are relatively independent since, on average, the p-values are the largest. Since the lower stock market capitalization of this sector, especially SPG, its volatility is almost independent of all other stocks. Finally, the volatilities of all the giant stocks (APPL, MSFT, JPM, KO, WMT) are highly dependent on others regardless of whether they are in the same sector or different. These findings are generally aligned with our intuition.

7. Conclusion

In this paper, we study the limiting behavior of the realized joint Laplace transform of volatilities by using non-overlapped and overlapped high-frequency data, including consistency and asymptotic normality. Moreover, we derive a central limit theorem for the joint Laplace transform under the assumption of the stationary volatility processes. Based on the results, we propose a consistent test for the dependence between two volatility processes. Simulation studies verify the finite sample performance of the proposed theory. We implement the test procedure using a real high-frequency data set.

This paper highlights some future research. For example, we may develop the large sample theory for asynchronous high-frequency data, consider the effect of market microstructure noise, study the related limiting behavior and independence tests for the multivariate volatilities, and consider a more general dependence structure of price processes, such as nonlinear dependence and so on.

† The R code is available upon request.

Table 3. *p*-value of the independence test for volatilities.

Sector	Information technology					Finance					Consumer staples					Real estate				
Symbol	APPL	MSFT	ORCL	IBM	ADBE	BAC	JPM	MS	WFC	PNC	KO	WMT	ADM	COST	KMB	O	SPG	KIM	CCL	WY
APPL		0.02	0.03	0.01	< 0.01	< 0.01	< 0.01	0.02	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	0.01	0.05	0.08	0.14	0.02	0.03	0.04
MSFT			0.02	0.03	0.11	0.02	< 0.01	0.01	0.03	0.04	< 0.01	< 0.01	< 0.01	0.01	0.30	< 0.01	0.11	0.01	< 0.01	0.03
ORCL				0.01	0.03	0.03	0.02	0.03	< 0.01	0.15	< 0.01	< 0.01	< 0.01	0.01	< 0.01	0.03	0.25	0.17	0.03	0.04
IBM					0.01	0.18	< 0.01	0.10	0.24	0.32	< 0.01	< 0.01	< 0.01	< 0.01	0.04	< 0.01	0.16	0.01	0.04	0.03
ADBE						0.17	0.01	0.04	0.32	0.13	0.03	0.03	< 0.01	0.04	0.22	0.11	0.43	0.08	0.03	< 0.01
BAC							< 0.01	0.08	0.18	0.04	< 0.01	0.01	< 0.01	< 0.01	< 0.01	0.01	0.11	< 0.01	0.01	0.01
JPM								0.04	0.21	< 0.01	< 0.01	< 0.01	0.01	< 0.01	< 0.01	< 0.01	0.02	< 0.01	< 0.01	0.02
MS									0.41	0.52	< 0.01	< 0.01	< 0.01	0.03	0.14	0.01	0.20	< 0.01	0.30	< 0.01
WFC										0.02	< 0.01	< 0.01	< 0.01	< 0.01	0.01	< 0.01	0.15	< 0.01	0.25	< 0.01
PNC											< 0.01	< 0.01	< 0.01	0.02	0.39	< 0.01	0.11	< 0.01	0.34	0.02
KO												< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	0.15	0.01	0.01	0.01
WMT													< 0.01	0.03	0.45	< 0.01	0.04	< 0.01	0.02	0.01
ADM														0.02	< 0.01	< 0.01	0.03	0.01	< 0.01	0.01
COST															0.55	0.33	0.70	0.18	0.04	0.45
KMB																< 0.01	0.60	< 0.01	0.09	< 0.01
O																	0.47	0.03	0.03	< 0.01
SPG																		0.04	0.78	0.13
KIM																			0.16	0.20
CCL																				0.55
WY																				

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Appendix. Proofs

This section contains all the proofs. Following usual conventions, we denote by C a generic constant that may change from line to line. We sometimes emphasize its dependence on some parameter p by writing C_p. Relying on the standard localization procedure, we can strengthen assumption 2.2 by assuming that all the processes studied in $[0, T]$ are bounded. For more details on the localization procedure, we refer to Jacod *et al.* (2019) or section 4.4.1 in Jacod and Protter (2012).

We denote

$$\alpha_i^Z := \alpha_{t_i^n}^Z, \quad \text{for } \alpha = b, \tilde{b}, v, v', \quad Z = X, Y, \quad \mathcal{G}_i = \mathcal{F}_{t_{2i}^n}, \quad i \geq 0.$$

Let

$$\xi_i^{(1)}(u,v) = 2\Delta_n^{\frac{1}{2}} \left\{ \cos\left(\frac{\sqrt{2u}\Delta_i^n X + \sqrt{2v}\Delta_{i+1}^n Y}{\sqrt{\Delta_n}} \right) - \cos\left(\frac{\sqrt{2u}\sigma_{i-1}^X \Delta_i^n W^X + \sqrt{2v}\sigma_{i-1}^Y \Delta_{i+1}^n W^Y}{\sqrt{\Delta_n}} \right) \right\},$$

$$\xi_i^{(2)}(u,v) = 2\Delta_n^{\frac{1}{2}} \left\{ \cos\left(\frac{\sqrt{2u}\sigma_{i-1}^X \Delta_i^n W^X + \sqrt{2v}\sigma_{i-1}^Y \Delta_{i+1}^n W^Y}{\sqrt{\Delta_n}} \right) - \mathrm{e}^{-\langle (u,v), ((\sigma_{i-1}^X)^2, (\sigma_{i-1}^Y)^2) \rangle} \right\},$$

$$\xi_i^{(3)}(u,v) = \Delta_n^{-\frac{1}{2}} \left\{ 2\Delta_n \mathrm{e}^{-\langle (u,v), ((\sigma_{i-1}^X)^2, (\sigma_{i-1}^Y)^2) \rangle} - \int_{t_{i-1}^n}^{t_{i+1}^n} \mathrm{e}^{-\langle (u,v), ((\sigma_s^X)^2, (\sigma_s^Y)^2) \rangle} \mathrm{d}s \right\}.$$

For every t, u and v, we can rewrite the left side of (4) as

$$\frac{1}{\sqrt{\Delta_n}} \left(V_n(u,v) - \int_0^T \mathrm{e}^{-\langle (u,v), ((\sigma_s^X)^2, (\sigma_s^Y)^2) \rangle} \mathrm{d}s \right) = \sum_{i=1}^{\lfloor n/2 \rfloor} \sum_{j=1}^{3} \xi_{2i-1}^{(j)}(u,v).$$

Before giving the proof of the main results, we first present the following lemma for the approximation error.

LEMMA A.1 *Let assumptions* 2.1–2.2 *hold, we have*

$$\sum_{i=1}^{\lfloor n/2 \rfloor} \xi_{2i-1}^{(1)}(u,v) \xrightarrow{P} 0, \quad \sum_{i=1}^{\lfloor n/2 \rfloor} \xi_{2i-1}^{(3)}(u,v) \xrightarrow{P} 0.$$

Proof First for $\sum_{i=1}^{\lfloor n/2 \rfloor} \xi_{2i-1}^{(3)}(u,v)$, note that $\xi_{2i-1}^{(3)}(u,v)$ can be written as $\sum_{j=1}^{3} \xi_{2i-1}^{(3,j)}(u,v)$, with

$$\xi_{2i-1}^{(3,1)}(u,v) = \Delta_n^{-\frac{1}{2}} \int_{t_{2i-2}^n}^{t_{2i}^n} \Big\langle k(\sigma_*^X, \sigma_*^Y, u, v) - k(\sigma_{2i-2}^X, \sigma_{2i-2}^Y, u, v), (\sigma_{2i-2}^X - \hat{\sigma}_s^X, \sigma_{2i-2}^Y - \hat{\sigma}_s^Y) \Big\rangle \mathrm{d}s,$$

$$\xi_{2i-1}^{(3,2)}(u,v) = \Delta_n^{-\frac{1}{2}} \int_{t_{2i-2}^n}^{t_{2i}^n} \Big\langle k(\sigma_{2i-2}^X, \sigma_{2i-2}^Y, u, v), (\sigma_{2i-2}^X - \hat{\sigma}_s^X, \sigma_{2i-2}^Y - \hat{\sigma}_s^Y) \Big\rangle \mathrm{d}s,$$

$$\xi_{2i-1}^{(3,3)}(u,v) = \Delta_n^{-\frac{1}{2}} \int_{t_{2i-2}^n}^{t_{2i}^n} \mathrm{e}^{\langle (u,v), ((\hat{\sigma}_s^X)^2, (\hat{\sigma}_s^Y)^2) \rangle} - \mathrm{e}^{\langle (u,v), ((\sigma_s^X)^2, (\sigma_s^Y)^2) \rangle} \mathrm{d}s,$$

where $k(x,y,u,v) = (-2ux\mathrm{e}^{-ux^2-vy^2}, -2vy\mathrm{e}^{-ux^2-vy^2})$, σ_*^Z is a number between σ_{2i-2}^Z and $\hat{\sigma}_s^Z$, $s \in [t_{2i-2}^n, t_{2i}^n]$, $Z = X, Y$, and

$$\hat{\sigma}_s^Z = \sigma_{2i-2}^Z + \int_{t_{2i-2}^n}^{s} v_r^Z \mathrm{d}W_r^Z + \int_{t_{2i-2}^n}^{s} v_r'^Z \mathrm{d}W_r'^Z + \int_{t_{2i-2}^n}^{s} \int_{\mathbb{R}} \delta'^Z(r-,z) \mu'^Z(\mathrm{d}r, \mathrm{d}z).$$

We denote the Euclidean space R^2 with the standard norm $\|(x_1,x_2)\| := \sqrt{x_1^2 + x_2^2}$, note that $\|k(x,y,u,v)\| \leq C_{u,v}$ and $\|k(x_1,y_1,u,v) - k(x_2,y_2,u,v)\| \leq C_{u,v}\sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$, for $x, y, x_1, x_2, y_1, y_2 \in \mathbb{R}$ and $u, v \geq 0$. For $\xi_{2i-1}^{(3,1)}(u,v)$, we have

$$\begin{aligned} &\mathbb{E}\left[\left| \xi_{2i-1}^{(3,1)}(u,v) \right| \right] \\ &\leq \Delta_n^{-\frac{1}{2}} \mathbb{E}\left[\sup_s \left\| k(\sigma_*^X, \sigma_*^Y, u, v) - k(\sigma_{2i-2}^X, \sigma_{2i-2}^Y, u, v) \right\| \times \int_{t_{2i-2}^n}^{t_{2i}^n} \left\| (\sigma_{2i-2}^X - \hat{\sigma}_s^X, \sigma_{2i-2}^Y - \hat{\sigma}_s^Y) \right\| \mathrm{d}s \right] \\ &\leq C_{u,v}\Delta_n^{-\frac{1}{2}} \sqrt{\mathbb{E}\left[\sup_s \left((\hat{\sigma}_s^X - \sigma_{2i-2}^X)^2 + (\hat{\sigma}_s^Y - \sigma_{2i-2}^Y)^2 \right) \right]} \\ &\quad \times \sqrt{2\Delta_n \mathbb{E}\left[\int_{t_{2i-2}^n}^{t_{2i}^n} \left\| (\sigma_{2i-2}^X - \hat{\sigma}_s^X, \sigma_{2i-2}^Y - \hat{\sigma}_s^Y) \right\|^2 \mathrm{d}s \right]}. \end{aligned} \quad \text{(A1)}$$

By the Burkholder–Davis–Gundy inequality and assumption 2.2, we have

$$\mathbb{E}\left[\sup_s (\hat{\sigma}_s^X - \sigma_{2i-2}^X)^2 \right]$$

$$\leq C\mathbb{E}\left[\int_{t_{2i-2}^n}^{t_{2i}^n}(v_r^X)^2\mathrm{d}r+\int_{t_{2i-2}^n}^{t_{2i}^n}(v_r^{'X})^2\mathrm{d}r\right.$$
$$\left.+\int_{t_{2i-2}^n}^{t_{2i}^n}\int_{\mathbb{R}}(\delta^{'X}(r-,x))^2\nu(\mathrm{d}x)\mathrm{d}r\right]$$
$$\leq C\Delta_n, \tag{A2}$$

and

$$\mathbb{E}\left[\sup_s(\hat{\sigma}_s^Y-\sigma_{2i-2}^Y)^2\right]\leq C\Delta_n. \tag{A3}$$

Therefore, it follows from (A1), (A2) and (A3) that

$$\mathbb{E}\left[\left|\xi_{2i-1}^{(3,1)}(u,v)\right|\right]\leq C_{u,v}\Delta_n^{3/2}. \tag{A4}$$

It is easy to check that

$$\mathbb{E}\left[\xi_{2i-1}^{(3,2)}(u,v)\middle|\mathcal{G}_{i-1}\right]=0, \tag{A5}$$

and

$$\mathbb{E}\left[\left|\xi_{2i-1}^{(3,2)}(u,v)\right|^2\right]$$
$$\leq C_{u,v}\mathbb{E}\left[\int_{t_{2i-2}^n}^{t_{2i}^n}\left\|(\sigma_{2i-2}^X-\hat{\sigma}_s^X,\sigma_{2i-2}^Y-\hat{\sigma}_s^Y)\right\|^2\mathrm{d}s\right]$$
$$\leq C_{u,v}\Delta_n^2. \tag{A6}$$

Applying the first-order Taylor expansion for $\xi_{2i-1}^{(3,3)}(u,v)$, we obtain

$$\mathbb{E}\left[\left|\xi_{2i-1}^{(3,3)}(u,v)\right|\middle|\mathcal{G}_{i-1}\right]$$
$$\leq C_{u,v}\Delta_n^{-\frac{1}{2}}\mathbb{E}\left[\int_{t_{2i-2}^n}^{t_{2i}^n}\left|(\hat{\sigma}_s^X)^2-(\sigma_s^X)^2\right|+\left|(\hat{\sigma}_s^Y)^2-(\sigma_s^Y)^2\right|\mathcal{G}_{i-1}\right].$$

Consequently, we have

$$\mathbb{E}\left[\left|\xi_{2i-1}^{(3,3)}(u,v)\right|\right]\leq C_{u,v}\Delta_n^{3/2}. \tag{A7}$$

Next, for $\sum_{i=1}^{\lfloor n/2\rfloor}\xi_{2i-1}^{(1)}(u,v)$, we can decompose $\xi_{2i-1}^{(1)}(u,v)$ as $\sum_{i=1}^{5}\xi_{2i-1}^{(1,i)}(u,v)$, with

$$\xi_{2i-1}^{(1,1)}(u,v)$$
$$=-4\Delta_n^{1/2}\sin\left(\frac{1}{2\sqrt{\Delta_n}}\sqrt{2u}\right.$$
$$\times\left(\Delta_{2i-1}^nX+\int_{t_{2i-2}^n}^{t_{2i-1}^n}b_s^X\mathrm{d}s+\int_{t_{2i-2}^n}^{t_{2i-1}^n}\sigma_s^X\mathrm{d}W_s^X\right)$$
$$\left.+\frac{1}{2\sqrt{\Delta_n}}\sqrt{2v}\left(\Delta_{2i}^nY+\int_{t_{2i-1}^n}^{t_{2i}^n}b_s^Y\mathrm{d}s+\int_{t_{2i-1}^n}^{t_{2i}^n}\sigma_s^Y\mathrm{d}W_s^Y\right)\right)$$
$$\times\sin\left(\frac{1}{2\sqrt{\Delta_n}}\sqrt{2u}\left(\Delta_{2i-1}^nX-\int_{t_{2i-2}^n}^{t_{2i-1}^n}b_s^X\mathrm{d}s\right.\right.$$
$$\left.-\int_{t_{2i-2}^n}^{t_{2i-1}^n}\sigma_s^X\mathrm{d}W_s^X\right)+\frac{1}{2\sqrt{\Delta_n}}\sqrt{2v}$$
$$\left.\times\left(\Delta_{2i}^nY-\int_{t_{2i-1}^n}^{t_{2i}^n}b_s^Y\mathrm{d}s-\int_{t_{2i-1}^n}^{t_{2i}^n}\sigma_s^Y\mathrm{d}W_s^Y\right)\right),$$

$$\xi_{2i-1}^{(1,2)}(u,v)$$
$$=-2\sqrt{2u}\Delta_n\sin\left(\frac{\sqrt{2u}\sigma_{2i-2}^X\Delta_{2i-1}^nW^X+\sqrt{2v}\sigma_{2i-2}^Y\Delta_{2i}^nW^Y}{\sqrt{\Delta_n}}\right)$$
$$\times b_{2i-2}^X-2\sqrt{2v}\Delta_n\sin$$
$$\times\left(\frac{\sqrt{2u}\sigma_{2i-2}^X\Delta_{2i-1}^nW^X+\sqrt{2v}\sigma_{2i-2}^Y\Delta_{2i}^nW^Y}{\sqrt{\Delta_n}}\right)b_{2i-1}^Y,$$

$$\xi_{2i-1}^{(1,3)}(u,v)$$
$$=-2\sin(\tilde{x}_1+\tilde{y}_1)\left(\sqrt{2u}\int_{t_{2i-2}^n}^{t_{2i-1}^n}(b_s^X-b_{2i-2}^X)\mathrm{d}s\right.$$
$$\left.+\sqrt{2v}\int_{t_{2i-1}^n}^{t_{2i}^n}(b_s^Y-b_{2i-1}^Y)\mathrm{d}s\right),$$

$$\xi_{2i-1}^{(1,4)}(u,v)$$
$$=-2\Delta_n^{-1/2}\cos$$
$$\times(\tilde{x}_2+\tilde{y}_2)\left(\sqrt{u}b_{2i-2}^X\Delta_n+\sqrt{u}\int_{t_{2i-2}^n}^{t_{2i-1}^n}(\sigma_s^X-\sigma_{2i-2}^X)\mathrm{d}W_s^X\right.$$
$$\left.+\sqrt{v}b_{2i-1}^Y\Delta_n+\sqrt{v}\int_{t_{2i-1}^n}^{t_{2i}^n}(\sigma_s^Y-\sigma_{2i-2}^Y)\mathrm{d}W_s^Y\right)^2,$$

$$\xi_{2i-1}^{(1,5)}(u,v)$$
$$=-2\sqrt{2u}\sin\left(\frac{\sqrt{2u}\sigma_{2i-2}^X\Delta_{2i-1}^nW^X+\sqrt{2v}\sigma_{2i-2}^Y\Delta_{2i}^nW^Y}{\sqrt{\Delta_n}}\right)$$
$$\times\int_{t_{2i-2}^n}^{t_{2i-1}^n}(\sigma_s^X-\sigma_{2i-2}^X)\mathrm{d}W_s^Y$$
$$-2\sqrt{2v}\sin\left(\frac{\sqrt{2u}\sigma_{2i-2}^X\Delta_{2i-1}^nW^X+\sqrt{2v}\sigma_{2i-2}^Y\Delta_{2i}^nW^Y}{\sqrt{\Delta_n}}\right)$$
$$\times\int_{t_{2i-1}^n}^{t_{2i}^n}(\sigma_s^Y-\sigma_{2i-2}^Y)\mathrm{d}W_s^Y,$$

where $\tilde{x}_1$ is between $\sqrt{2u}\Delta_n^{-1/2}\int_{t_{2i-2}^n}^{t_{2i-1}^n}b_s^X\mathrm{d}s+\sqrt{2u}\Delta_n^{-1/2}\int_{t_{2i-2}^n}^{t_{2i-1}^n}\sigma_s^X\mathrm{d}W_s^X$ and $\sqrt{2u}\Delta_n^{-1/2}b_{2i-2}^X\Delta_n+\int_{t_{2i-2}^n}^{t_{2i-1}^n}\sigma_s^X\mathrm{d}W_s^X$;

$\tilde{y}_1$ is between $\sqrt{2v}\Delta_n^{-1/2}\int_{t_{2i-1}^n}^{t_{2i}^n}b_s^Y\mathrm{d}s+\sqrt{2v}\Delta_n^{-1/2}\int_{t_{2i-1}^n}^{t_{2i}^n}\sigma_s^Y\mathrm{d}W_s^Y$ and $\sqrt{2v}\Delta_n^{-1/2}b_{2i-1}^Y\Delta_n+\int_{t_{2i-1}^n}^{t_{2i}^n}\sigma_s^Y\mathrm{d}W_s^Y$;

$\tilde{x}_2$ is between $\sqrt{2u}\Delta_n^{-1/2}b_{2i-2}^X\Delta_n+\int_{t_{2i-2}^n}^{t_{2i-1}^n}\sigma_s^X\mathrm{d}W_s^X$ and $\sqrt{2u}\Delta_n^{-1/2}\sigma_{2i-2}^X\Delta_{2i-1}^nW^X$;

$\tilde{y}_2$ is between $\sqrt{2v}\Delta_n^{-1/2}b_{2i-1}^Y\Delta_n+\int_{t_{2i-1}^n}^{t_{2i}^n}\sigma_s^Y\mathrm{d}W_s^Y$ and $\sqrt{2v}\Delta_n^{-1/2}\sigma_{2i-2}^Y\Delta_{2i}^nW^Y$.

Following the inequalities $|\sin(A+B)|\leq|\sin(A)|+|\sin(B)|$, $|\sin(x)|\leq|x|$, we have

$$\mathbb{E}\left[\left|\xi_{2i-1}^{(1,1)}(u,v)\right|\right]$$
$$\leq C_{u,v}\Delta_n^{1/2}\left\{\left|\sin\left(\frac{1}{2\sqrt{\Delta_n}}\sqrt{2u}\int_{t_{2i-2}^n}^{t_{2i-1}^n}\int_{\mathbb{R}}\delta^X(s-,x)\mu^X(\mathrm{d}s,\mathrm{d}x)\right)\right|\right.$$
$$\left.+\left|\sin\left(\frac{1}{2\sqrt{\Delta_n}}\sqrt{2u}\int_{t_{2i-1}^n}^{t_{2i}^n}\int_{\mathbb{R}}\delta^Y(s-,y)\mu^Y(\mathrm{d}s,\mathrm{d}y)\right)\right|\right\}$$
$$\leq C_{u,v}\Delta_n^{1/2-\beta/2-\iota/2}\left\{\mathbb{E}\left|\int_{t_{2i-2}^n}^{t_{2i-1}^n}\int_{\mathbb{R}}\delta^X(s-,x)\mu^X(\mathrm{d}s,\mathrm{d}x)\right|^{\beta+\iota}\right.$$
$$\left.+\mathbb{E}\left|\int_{t_{2i-1}^n}^{t_{2i}^n}\int_{\mathbb{R}}\delta^Y(s-,y)\mu^Y(\mathrm{d}s,\mathrm{d}y)\right|^{\beta+\iota}\right\}$$

$$\leq C_{u,v}\Delta_n^{1/2-\beta/2-\iota/2}\left\{\mathbb{E}\int_{t_{2i-2}^n}^{t_{2i-1}^n}\int_{\mathbb{R}}\left|\delta^X(s-,x)\right|^{\beta+\iota}\nu^X(\mathrm{d}x)\mathrm{d}s\right.$$

$$\left.+\,\mathbb{E}\int_{t_{2i-1}^n}^{t_{2i}^n}\int_{\mathbb{R}}\left|\delta^Y(s-,y)\right|^{\beta+\iota}\nu^Y(\mathrm{d}y)\mathrm{d}s\right\} \tag{A8}$$

$$\leq C_{u,v}\Delta_n^{3/2-\beta/2-\iota/2}, \tag{A9}$$

where $\iota\in(0,1-\beta)$. The second last inequality is C_r-inequality $\mathbb{E}|\sum_{i=1}^n X_i|^r\leq C_r\sum_{i=1}^n\mathbb{E}|X_i|^r$, when $0<r\leq 1$, $C_r=1$; when $r>1$, $C_r=n^{r-1}$. Therefore, if the price process has infinite variation jump, namely $\beta+\iota>1$. We cannot guarantee $\xi_{2i-1}^{(1,1)}(u,v)=o_p(\Delta_n)$.

Applying the same techniques, we have

$$\mathbb{E}\left[\xi_{2i-1}^{(1,2)}(u,v)\middle|\mathcal{G}_{n-1}\right]=0;\quad \mathbb{E}\left[\left|\xi_{2i-1}^{(1,2)}(u,v)\right|^2\right]\leq C_{u,v}\Delta_n^2, \tag{A10}$$

$$\mathbb{E}\left[\left|\xi_{2i-1}^{(1,3)}(u,v)\right|\right]\leq C_{u,v}\Delta_n^{3/2}, \tag{A11}$$

$$\mathbb{E}\left[\left|\xi_{2i-1}^{(1,4)}(u,v)\right|\right]\leq C_{u,v}\Delta_n^{3/2}. \tag{A12}$$

For the remaining term $\xi_{2i-1}^{(1,5)}(u,v)$, we decompose it as $\sum_{i=1}^2\xi_{2i-1}^{(1,5,i)}(u,v)$, with

$$\xi_{2i-1}^{(1,5,1)}(u,v)=-2f(\Delta_{2i-1}^nW^X,\Delta_{2i}^nW^Y)\times\left(\sqrt{2u}\int_{t_{2i-2}^n}^{t_{2i-1}^n}\zeta_s^X\mathrm{d}W_s^X+\sqrt{2v}\int_{t_{2i-1}^n}^{t_{2i}^n}\zeta_s^Y\mathrm{d}W_s^Y\right),$$

$$\xi_{2i-1}^{(1,5,2)}(u,v)=-2f(\Delta_{2i-1}^nW^X,\Delta_{2i}^nW^Y)\times\left(\sqrt{2u}\int_{t_{2i-2}^n}^{t_{2i-1}^n}\eta_s^X\mathrm{d}W_s^X+\sqrt{2v}\int_{t_{2i-1}^n}^{t_{2i}^n}\eta_s^Y\mathrm{d}W_s^Y\right),$$

where

$$\zeta_s^Z=\int_{t_{2i-2}^n}^s\nu_{2i-2}^Z\mathrm{d}W_r^Z+\int_{t_{2i-2}^n}^s\nu_{2i-2}^{\prime Z}\mathrm{d}W_r^{\prime Z}+\int_{t_{2i-2}^n}^s\int_{\mathbb{R}}\delta^{\prime Z}(t_{2i-2}^n-,z)\mu^{\prime Z}(\mathrm{d}r,\mathrm{d}z),$$

$$\eta_s^Z=\int_{t_{2i-2}^n}^s\tilde{b}_r^Z\mathrm{d}r+\int_{t_{2i-2}^n}^s\left(\nu_r^Z-\nu_{2i-2}^Z\right)\mathrm{d}W_r^Z+\int_{t_{2i-2}^n}^s\left(\nu_r^{\prime Z}-\nu_{2i-2}^{\prime Z}\right)\mathrm{d}W_r^{\prime Z}+\int_{t_{2i-2}^n}^s\int_{\mathbb{R}}\left(\delta^{\prime Z}(r-,z)-\delta^{\prime Z}(t_{2i-2}^n-,z)\right)\mu^{\prime Z}(\mathrm{d}r,\mathrm{d}z),\quad Z=X,Y,$$

$$f(x,y)=\sin\left(\frac{\sqrt{2u}\sigma_{2i-2}^Xx+\sqrt{2v}\sigma_{2i-2}^Yy}{\sqrt{\Delta_n}}\right).$$

Applying Itô's formula, we have

$$\mathbb{E}\left[f(\Delta_{2i-1}^nW^X,\Delta_{2i}^nW^Y)\int_{t_{2i-2}^n}^{t_{2i-1}^n}\times\left(\zeta_s^X-\int_{t_{2i-2}^n}^s\nu_{2i-2}^X\mathrm{d}W_r^X\right)\mathrm{d}W_s^X\middle|\mathcal{G}_{i-1}\right]=0$$

and

$$\mathbb{E}\left[f(\Delta_{2i-1}^nW^X,\Delta_{2i}^nW^Y)\int_{t_{2i-2}^n}^{t_{2i-1}^n}\left(W_s^X-W_{2i-2}^X\right)\mathrm{d}W_s^X\middle|\mathcal{G}_{i-1}\right]=\frac{1}{2}\mathbb{E}\left[f(\Delta_{2i-1}^nW^X,\Delta_{2i}^nW^Y)\left(\left(\Delta_{2i-1}^nW^X\right)^2-\Delta_n\right)\middle|\mathcal{G}_{i-1}\right]=0.$$

Similarly, we obtain

$$\mathbb{E}\left[\xi_{2i-1}^{(1,5,1)}(u,v)\middle|\mathcal{G}_{i-1}\right]=0;\quad\mathbb{E}\left[\left|\xi_{2i-1}^{(1,5,1)}(u,v)\right|^2\right]\leq C_{u,v}\Delta_n^2, \tag{A13}$$

$$\mathbb{E}\left[\left|\xi_{2i-1}^{(1,5,2)}(u,v)\right|\right]\leq C_{u,v}\Delta_n^{3/2}. \tag{A14}$$

Finally, it follows from (A4) to (A14), that

$$\sum_{i=1}^{\lfloor n/2\rfloor}\xi_{2i-1}^{(1)}(u,v)\xrightarrow{P}0,\qquad\sum_{i=1}^{\lfloor n/2\rfloor}\xi_{2i-1}^{(3)}(u,v)\xrightarrow{P}0.$$

■

Now, we give proof of the main results.

Proof of Theorem 3.1. By lemma A.1, we have

$$\sum_{i=1}^{\lfloor n/2\rfloor}\xi_{2i-1}^{(1)}(u,v)\xrightarrow{P}0,\qquad\sum_{i=1}^{\lfloor n/2\rfloor}\xi_{2i-1}^{(3)}(u,v)\xrightarrow{P}0.$$

Let $\zeta_i^{(2)}(u,v)=\xi_{2i-1}^{(2)}(u,v)$ and note that $\{\zeta_j^{(2)}(u,v),\ j=1,2,\ldots,\lfloor n/2\rfloor\}$ is a $\mathcal{G}$-martingale sequence. The argument used in the proof of theorem 1 of Todorov and Tauchen (2012a) shows that we need to prove the finite-dimensional convergence of $\sum_{j=1}^{\lfloor n/2\rfloor}\zeta_j^{(2)}(u,v)$ and the tightness of $\sum_{i=1}^{\lfloor n/2\rfloor}\sum_{j=1}^3\xi_{2i-1}^{(j)}(u,v)$ to complete the proof of the stable convergence in (4).

By a standard argument of the stable limit theorem, as stated in theorem IX.7.28 of Jacod and Shiryaev (2003), it is enough to show the following results for the finite-dimensional convergence of $\sum_{j=1}^{\lfloor n/2\rfloor}\zeta_j^{(2)}(u,v)$:

$$\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\left[\zeta_j^{(2)}(u,v)\zeta_j^{(2)}(u',v')\middle|\mathcal{G}_{j-1}\right]\xrightarrow{P}F_p(u,v,u',v'), \tag{A15}$$

$$\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\left[\left(\zeta_j^{(2)}(u,v)\right)^4\middle|\mathcal{G}_{j-1}\right]\xrightarrow{P}0, \tag{A16}$$

$$\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\left[\zeta_j^{(2)}(u,v)\delta_jW^Z\middle|\mathcal{G}_{j-1}\right]\xrightarrow{P}0,\quad\text{for }Z=X,Y, \tag{A17}$$

$$\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\left[\zeta_j^{(2)}(u,v)\delta_jN\middle|\mathcal{G}_{j-1}\right]\xrightarrow{P}0, \tag{A18}$$

where N is any bounded $\mathcal{G}$-martingale orthogonal to W^X and W^Y, $\delta_jW^Z=W_{2j\Delta_n}^Z-W_{2(j-1)\Delta_n}^Z$ and $\delta_jN=N_{2j\Delta_n}-N_{2(j-1)\Delta_n}$.

Theorem IX.7.28 of Jacod and Shiryaev (2003) employs the conditional Lindeberg condition ((7.30) of theorem IX.7.28) but we use Lyapunov condition (A16) to prove central limit theorem. Because the latter is the sufficient condition for Lindeberg condition. Moreover, the Lyapunov condition is easier to verify than the Lindeberg condition.

To show (A15), we have

$$\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\left[\zeta_j^{(2)}(u,v)\zeta_j^{(2)}(u',v')\middle|\mathcal{G}_{j-1}\right]$$

$$=2\Delta_n\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\left[\cos\left(\frac{(\sqrt{2u}+\sqrt{2u'})\sigma_{2j-2}^X\Delta_{2j-1}^nW^X+(\sqrt{2v}+\sqrt{2v'})\sigma_{2j-2}^Y\Delta_{2j}^nW^Y}{\sqrt{\Delta_n}}\right)\middle|\mathcal{G}_{j-1}\right]$$

$$+\,2\Delta_n\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\left[\cos\left(\frac{\begin{array}{c}(\sqrt{2u}-\sqrt{2u'})\sigma^X_{2j-2}\Delta^n_{2j-1}W^X\\ +(\sqrt{2v}-\sqrt{2v'})\sigma^Y_{2j-2}\Delta^n_{2j}W^Y\end{array}}{\sqrt{\Delta_n}}\right)\middle|\,\mathcal{G}_{j-1}\right]$$

$$-\,4\Delta_n\sum_{j=1}^{\lfloor n/2\rfloor}\mathrm{e}^{-\langle(u+u',v+v'),((\sigma^X_{2j-2})^2,(\sigma^Y_{2j-2})^2)\rangle}$$

$$=2\Delta_n\sum_{j=1}^{\lfloor n/2\rfloor}\Big\{\mathrm{e}^{-(\sqrt{u}+\sqrt{u'})^2(\sigma^X_{2j-2})^2-(\sqrt{v}+\sqrt{v'})^2(\sigma^Y_{2j-2})^2}$$

$$+\mathrm{e}^{-(\sqrt{u}-\sqrt{u'})(\sigma^X_{2j-2})^2-(\sqrt{v}-\sqrt{v'})(\sigma^Y_{2j-2})^2}\Big\}$$

$$-\,4\Delta_n\sum_{i=1}^{\lfloor n/2\rfloor}\mathrm{e}^{-\langle(u+u',v+v'),((\sigma^X_{2j-2})^2,(\sigma^Y_{2j-2})^2)\rangle}$$

$$\xrightarrow{P}\int_0^T\Big\{\mathrm{e}^{-(\sqrt{u}+\sqrt{u'})^2(\sigma^X_s)^2-(\sqrt{v}+\sqrt{v'})^2(\sigma^Y_s)^2}$$

$$+\mathrm{e}^{-(\sqrt{u}-\sqrt{u'})^2(\sigma^X_s)^2-(\sqrt{v}-\sqrt{v'})^2(\sigma^Y_s)^2}$$

$$-2\mathrm{e}^{-(u+u')(\sigma^X_s)^2-(v+v')(\sigma^Y_s)^2}\Big\}\,\mathrm{d}s.$$

For (A16), we simply have

$$\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\left[\left(\zeta^{(2)}_j(u,v)\right)^4\middle|\,\mathcal{G}_{j-1}\right]\le C\Delta_n\xrightarrow{P}0.$$

Next, we show (A17) by letting $Z=X$, since the proof for $Z=Y$ is similar. Observing that

$$\begin{aligned}
&\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\left[\zeta^{(2)}_j(u,v)\delta_jW^X\middle|\,\mathcal{G}_{j-1}\right]\\
&=2\Delta_n^{\frac12}\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\left[\cos\left(\frac{\sqrt{2u}\sigma^X_{2j-2}\Delta^n_{2j-1}W^X+\sqrt{2v}\sigma^Y_{2j-2}\Delta^n_{2j}W^Y}{\sqrt{\Delta_n}}\right)\delta_jW^X\middle|\,\mathcal{G}_{j-1}\right]\\
&=2\Delta_n^{\frac12}\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\left[\cos\left(\frac{\sqrt{2u}\sigma^X_{2j-2}\Delta^n_{2j-1}W^X+\sqrt{2v}\sigma^Y_{2j-2}\Delta^n_{2j}W^Y}{\sqrt{\Delta_n}}\right)\Delta^n_{2j-1}W^X\middle|\,\mathcal{G}_{j-1}\right]\\
&\quad+2\Delta_n^{\frac12}\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\left[\cos\left(\frac{\sqrt{2u}\sigma^X_{2j-2}\Delta^n_{2j-1}W^X+\sqrt{2v}\sigma^Y_{2j-2}\Delta^n_{2j}W^Y}{\sqrt{\Delta_n}}\right)\Delta^n_{2j}W^X\middle|\,\mathcal{G}_{j-1}\right],
\end{aligned}\tag{A19}$$

the first term of (A19) is equal to zero because the variable inside the expectation is an odd function of $\Delta^n_{2j-1}W^X$ and $\Delta^n_{2j}W^Y$ is independent of $\Delta^n_{2j-1}W^X$. For the second term of (A19), we have

$$\begin{aligned}
&\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\left[\cos\left(\frac{\begin{array}{c}\sqrt{2u}\sigma^X_{2j-2}\Delta^n_{2j-1}W^X\\ +\sqrt{2v}\sigma^Y_{2j-2}\Delta^n_{2j}W^Y\end{array}}{\sqrt{\Delta_n}}\right)\Delta^n_{2j}W^X\middle|\,\mathcal{G}_{j-1}\right]\\
&=\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\Bigg[\cos\left(\frac{\sqrt{2u}\sigma^X_{2j-2}\Delta^n_{2j-1}W^X}{\sqrt{\Delta_n}}\right)\\
&\qquad\times\cos\left(\frac{\sqrt{2v}\sigma^Y_{2j-2}\Delta^n_{2j}W^Y}{\sqrt{\Delta_n}}\right)\Delta^n_{2j}W^X\Bigg|\,\mathcal{G}_{j-1}\Bigg]\\
&\quad-\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\Bigg[\sin\left(\frac{\sqrt{2u}\sigma^X_{2j-2}\Delta^n_{2j-1}W^X}{\sqrt{\Delta_n}}\right)\\
&\qquad\times\sin\left(\frac{\sqrt{2v}\sigma^Y_{2j-2}\Delta^n_{2j}W^Y}{\sqrt{\Delta_n}}\right)\Delta^n_{2j}W^X\Bigg|\,\mathcal{G}_{j-1}\Bigg].
\end{aligned}\tag{A20}$$

The second term of (A20) is zero because $\sin(\cdot)$ is an odd function and $\Delta^n_{2j-1}W^X$ is independent of both $\Delta^n_{2j}W^X$ and $\Delta^n_{2j}W^Y$. For the first term of (A20), Itô's formula yields

$$\begin{aligned}
&\cos\left(\frac{\sqrt{2v}\sigma^Y_{2j-2}\Delta^n_{2j}W^Y}{\sqrt{\Delta_n}}\right)\\
&=1+\int_{t^n_{2j-1}}^{t^n_{2j}}\sin\left(\frac{\sqrt{2v}\sigma^Y_{2j-2}\int_{t^n_{2j-1}}^s\mathrm{d}W^Y_u}{\sqrt{\Delta_n}}\right)\frac{\sqrt{2v}}{\sqrt{\Delta_n}}\sigma^Y_{2j-2}\mathrm{d}W^Y_s\\
&\quad-\frac12\int_{t^n_{2j-1}}^{t^n_{2j}}\cos\left(\frac{\sqrt{2v}\sigma^X_{2j-2}\int_{t^n_{2j-1}}^s\mathrm{d}W^Y_u}{\sqrt{\Delta_n}}\right)\left(\frac{\sqrt{2v}}{\sqrt{\Delta_n}}\sigma^Y_{2j-2}\right)^2\mathrm{d}s.
\end{aligned}$$

Hence, by the product formula, we obtain

$$\begin{aligned}
&\mathbb{E}\left[\cos\left(\frac{\sqrt{2v}\sigma^Y_{2j-2}\Delta^n_{2j}W^Y}{\sqrt{\Delta_n}}\right)\Delta^n_{2j}W^X\middle|\,\mathcal{G}_{j-1}\right]\\
&=\mathbb{E}\Bigg[\int_{t^n_{2j-1}}^{t^n_{2j}}\cos\left(\frac{\sqrt{2v}\sigma^X_{2j-2}\int_{t^n_{2j-1}}^s\mathrm{d}W^Y_u}{\sqrt{\Delta_n}}\right)\\
&\qquad\times\left(\frac{\sqrt{2v}}{\sqrt{\Delta_n}}\sigma^Y_{2j-2}\right)^2\int_{t^n_{2j-1}}^s\mathrm{d}W^X_u\,\mathrm{d}s\Bigg|\,\mathcal{G}_{j-1}\Bigg].
\end{aligned}$$

Now, we define a function

$$f(s):=\mathbb{E}\left[\cos\left(\frac{\sqrt{2v}\sigma^Y_{2j-2}\int_{t^n_{2j-1}}^s\mathrm{d}W^Y_u}{\sqrt{\Delta_n}}\right)\int_{t^n_{2j-1}}^s\mathrm{d}W^X_u\middle|\,\mathcal{G}_{j-1}\right]$$

for $s\in[t^n_{2j-1},t^n_{2j}]$, then we have $f(t^n_{2j-1})=0$ and $f(s)=c\int_{t^n_{2j-1}}^s f(u)$ *du* where c is a constant. It is easy to see that $f(s)$ is continuous, then $f(s)\equiv 0$ in view of the initial condition. Therefore, we have

$$\begin{aligned}
&\sum_{j=1}^{\lfloor n/2\rfloor}\mathbb{E}\Bigg[\cos\left(\frac{\sqrt{2u}\sigma^X_{2j-2}\Delta^n_{2j-1}W^X}{\sqrt{\Delta_n}}\right)\\
&\qquad\times\cos\left(\frac{\sqrt{2v}\sigma^Y_{2j-2}\Delta^n_{2j}W^Y}{\sqrt{\Delta_n}}\right)\Delta^n_{2j}W^X\Bigg|\,\mathcal{G}_{j-1}\Bigg]
\end{aligned}$$

$$= \sum_{j=1}^{\lfloor n/2 \rfloor} \mathbb{E}\left[\cos\left(\frac{\sqrt{2u}\sigma^X_{2j-2}\Delta^n_{2j-1}W^X}{\sqrt{\Delta_n}}\right)\middle| \mathcal{G}_{j-1}\right]$$
$$\times \mathbb{E}\left[\int_{t^n_{2j-2}}^{t^n_{2j}} \cos\left(\frac{\sqrt{2v}\sigma^X_{2j-2}\int_{t^n_{2j}}^s \mathrm{d}W^Y_u}{\sqrt{\Delta_n}}\right)\right.$$
$$\left.\times \left(\frac{\sqrt{2v}}{\sqrt{\Delta_n}}\sigma^Y_{2j-2}\right)^2 \int_{t^n_{2j}}^s \mathrm{d}W^X_u \,\mathrm{d}s \middle| \mathcal{G}_{j-1}\right]$$
$$= 0.$$

Finally, for (A18), since N is orthogonal to W^X and W^Y and defined on $\mathcal{F}$, we have

$$\mathbb{E}\left[\zeta^{(2)}_j(u,v)\delta_j N\middle| \mathcal{G}_{j-1}\right] = 0,$$

which yields the desired result.

We now show the tightness. For the convenience of notations, we denote

$$\hat{S}_{t,1}(u,v) = \sum_{i=\lfloor (t-1)/2\Delta_n\rfloor+1}^{\lfloor t/2\Delta_n\rfloor} \xi^{(2)}_{2i-1}(u,v),$$
$$\hat{S}_{t,2}(u,v) = \sum_{i=\lfloor (t-1)/2\Delta_n\rfloor+1}^{\lfloor t/2\Delta_n\rfloor} \left(\xi^{(3,2)}_{2i-1}(u,v) + \xi^{(1,2)}_{2i-1}(u,v) + \xi^{(1,5,1)}_{2i-1}(u,v)\right),$$
$$\hat{S}_{t,3}(u,v) = \sum_{i=\lfloor (t-1)/2\Delta_n\rfloor+1}^{\lfloor t/2\Delta_n\rfloor} \left(\xi^{(1)}_{2i-1}(u,v) + \xi^{(3)}_{2i-1}(u,v)\right) - \hat{S}_{t,2}(u,v).$$

For $u_1, u_2, v_1, v_2 \in \mathbb{R}_+$,

$$\mathbb{E}\left[\sum_{t=1}^T \left(\hat{S}_{t,1}(u_2,v_2) + \hat{S}_{t,2}(u_2,v_2)\right)\right]^2 \le C, \tag{A21}$$

$$\mathbb{E}\left[\sum_{t=1}^T \left(\hat{S}_{t,1}(u_1,v_1) + \hat{S}_{t,2}(u_1,v_1) - \hat{S}_{t,1}(u_2,v_2) - \hat{S}_{t,2}(u_2,v_2)\right)\right]^2$$
$$\le C\left[(\sqrt{u_1}-\sqrt{u_2})^2 + (\sqrt{v_1}-\sqrt{v_2})^2\right] \vee \left[(u_1-u_2)^2 + (v_1-v_2)^2\right]. \tag{A22}$$

According to theorem 20 of Ibragimov and Has' Minskii (1981), we have $\sum_{t=1}^T(\hat{S}_{t,1}(u,v) + \hat{S}_{t,2}(u,v))$ is tight. Since

$$\Delta_n^{\beta/2+\iota/2-1/2}\left|\sup_{0\le u\le\bar{u}, 0\le v\le\bar{v}} \sum_{i=1}^{\lfloor T/2\Delta_n\rfloor} \hat{S}_{t,3}(u,v)\right|$$

is bounded in probability. Therefore, $\sum_{i=1}^{\lfloor n/2\rfloor}\sum_{j=1}^3 \xi^{(j)}_{2i-1}(u,v)$ is tight.

Finally, we prove that the $\tilde{\Gamma}_n$ is a consistent estimator:

$$\tilde{\Gamma}_n = 4\Delta_n \sum_{i=1}^{\lfloor n/2\rfloor} \cos\left(\frac{\sqrt{2u}\Delta^n_{2i-1}X + \sqrt{2v}\Delta^n_{2i}Y}{\sqrt{\Delta_n}}\right) \times \cos\left(\frac{\sqrt{2u'}\Delta^n_{2i-1}X + \sqrt{2v'}\Delta^n_{2i}Y}{\sqrt{\Delta_n}}\right)$$
$$-4\Delta_n \sum_{i=1}^{\lfloor n/2\rfloor} \cos\left(\frac{\sqrt{2(u+u')}\Delta^n_{2i-1}X + \sqrt{2(v+v')}\Delta^n_{2i}Y}{\sqrt{\Delta_n}}\right).$$

$$= 4\sum_{i=1}^{\lfloor n/2\rfloor} \frac{1}{2}\Delta_n\left[\cos\left(\frac{(\sqrt{2u}+\sqrt{2u'})\Delta^n_{2i-1}X + (\sqrt{2v}+\sqrt{2v'})\Delta^n_{2i}Y}{\sqrt{\Delta_n}}\right)\right.$$
$$\left.+\cos\left(\frac{(\sqrt{2u}-\sqrt{2u'})\Delta^n_{2i-1}X + (\sqrt{2v}-\sqrt{2v'})\Delta^n_{2i}Y}{\sqrt{\Delta_n}}\right)\right]$$
$$-4\sum_{i=1}^{\lfloor n/2\rfloor} \Delta_n \cos\left(\frac{\sqrt{2(u+u')}\Delta^n_{2i-1}X + \sqrt{2(v+v')}\Delta^n_{2i}Y}{\sqrt{\Delta_n}}\right).$$

By the Weak Law of Large Numbers,

$$\tilde{\Gamma}_n - \left[2\Delta_n \sum_{i=1}^{\lfloor n/2\rfloor} e^{-(\sqrt{u}+\sqrt{u'})^2(\sigma^X_{2i-2})^2 - (\sqrt{v}+\sqrt{v'})^2(\sigma^Y_{2i-2})^2}\right.$$
$$+2\Delta_n \sum_{i=1}^{\lfloor n/2\rfloor} e^{-(\sqrt{u}-\sqrt{u'})^2(\sigma^X_{2i-2})^2 - (\sqrt{v}-\sqrt{v'})^2(\sigma^Y_{2i-2})^2}$$
$$\left.-4\Delta_n \sum_{i=1}^{\lfloor n/2\rfloor} e^{-(u+u')(\sigma^X_{2i-2})^2 - (v+v')(\sigma^Y_{2i-2})^2}\right] \xrightarrow{P} 0$$

and

$$2\Delta_n \sum_{i=1}^{\lfloor n/2\rfloor} e^{-(\sqrt{u}+\sqrt{u'})^2(\sigma^X_{2i-2})^2 - (\sqrt{v}+\sqrt{v'})^2(\sigma^Y_{2i-2})^2}$$
$$+ 2\Delta_n \sum_{i=1}^{\lfloor n/2\rfloor} e^{-(\sqrt{u}-\sqrt{u'})^2(\sigma^X_{2i-2})^2 - (\sqrt{v}-\sqrt{v'})^2(\sigma^Y_{2i-2})^2}$$
$$- 4\Delta_n \sum_{i=1}^{\lfloor n/2\rfloor} e^{-(u+u')(\sigma^X_{2i-2})^2 - (v+v')(\sigma^Y_{2i-2})^2}$$
$$\xrightarrow{P} \int_0^T \left\{\mathrm{e}^{-(\sqrt{u}+\sqrt{u'})^2(\sigma^X_s)^2 - (\sqrt{v}+\sqrt{v'})^2(\sigma^Y_s)^2} + \mathrm{e}^{-(\sqrt{u}-\sqrt{u'})^2(\sigma^X_s)^2 - (\sqrt{v}-\sqrt{v'})^2(\sigma^Y_s)^2} - 2\mathrm{e}^{-(u+u')(\sigma^X_s)^2 - (v+v')(\sigma^Y_s)^2}\right\}\mathrm{d}s.$$

We obtain

$$\tilde{\Gamma}_n \xrightarrow{P} \int_0^T \left\{\mathrm{e}^{-(\sqrt{u}+\sqrt{u'})^2(\sigma^X_s)^2 - (\sqrt{v}+\sqrt{v'})^2(\sigma^Y_s)^2} + \mathrm{e}^{-(\sqrt{u}-\sqrt{u'})^2(\sigma^X_s)^2 - (\sqrt{v}-\sqrt{v'})^2(\sigma^Y_s)^2} - 2\mathrm{e}^{-(u+u')(\sigma^X_s)^2 - (v+v')(\sigma^Y_s)^2}\right\}\mathrm{d}s.$$

■

Proof of theorem 3.2. We redefine

$$\xi^{(1)}_i(u,v) = \Delta_n^{\frac{1}{2}}\left\{\cos\left(\frac{\sqrt{2u}\Delta^n_i X + \sqrt{2v}\Delta^n_{i+1}Y}{\sqrt{\Delta_n}}\right) - \cos\left(\frac{\sqrt{2u}\sigma^X_{i-1}\Delta^n_i W^X + \sqrt{2v}\sigma^Y_{i-1}\Delta^n_{i+1}W^Y}{\sqrt{\Delta_n}}\right)\right\},$$

$$\xi^{(2)}_i(u,v) = \Delta_n^{\frac{1}{2}}\left\{\cos\left(\frac{\sqrt{2u}\sigma^X_{i-1}\Delta^n_i W^X + \sqrt{2v}\sigma^Y_{i-1}\Delta^n_{i+1}W^Y}{\sqrt{\Delta_n}}\right) - \mathrm{e}^{-\langle (u,v),((\sigma^X_{i-1})^2,(\sigma^Y_{i-1})^2)\rangle}\right\},$$

$$\xi^{(3)}_i(u,v) = \Delta_n^{-\frac{1}{2}}\left\{\Delta_n \mathrm{e}^{-\langle (u,v),((\sigma^X_{i-1})^2,(\sigma^Y_{i-1})^2)\rangle}\right.$$

$$-\int_{(i-1)\Delta_n}^{i\Delta_n} \mathrm{e}^{-\langle (u,v),((\sigma_s^X)^2,(\sigma_s^Y)^2)\rangle}\mathrm{d}s\bigg\}.$$

By the same techniques used in the proof of lemma A.1, we can show that

$$\sum_{i=1}^{n-1}\xi_i^{(1)}(u,v)\xrightarrow{P}0,\quad \sum_{i=1}^{n-1}\xi_i^{(3)}(u,v)\xrightarrow{P}0. \tag{A23}$$

Similar to the proof of theorem 3.1, to obtain the convergence in (5), we only need to show the finite-dimensional convergence of $\sum_{i=1}^{n-1}\xi_i^{(2)}(u,v)$ and the tightness of $\sum_{i=1}^{n-1}\sum_{j=1}^{3}\xi_i^{(j)}(u,v)$.

We first consider $\sum_{i=1}^{n-1}\xi_i^{(2)}(u,v)$. To deal with the dependence of increments, we use the big-small-block method. Finally, those big blocks dominate the asymptotic performance, while those small blocks are asymptotically negligible. To create, we first split the increments so that each block contains $p+1$ increments, and then gather the first p increments (big block), and the last one belongs to another block (small block). Precisely, let $N_p=\lfloor\frac{n-1}{p+1}\rfloor$, $\alpha_j^p:=(j-1)(p+1)$, and

$$\zeta_j^{n,p}(u,v)=\sum_{i=\alpha_j^p+1}^{\alpha_{j+1}^p-1}\xi_i^{(2)}(u,v),\quad \eta_j^{n,p}(u,v)=\xi_{\alpha_{j+1}^p}^{(2)}(u,v).$$

Hence,

$$\sum_{i=1}^{n-1}\xi_i^{(2)}(u,v)=\sum_{j=1}^{N_p}\zeta_j^{n,p}(u,v)+\sum_{j=1}^{N_p}\eta_j^{n,p}(u,v)+\sum_{i=N_p(p+1)+1}^{n-1}\xi_i^{(2)}(u,v).$$

Observing that $\sup_{u,v}\sup_i|\xi_i^{(2)}(u,v)|\le 2\Delta_n^{1/2}$, thus

$$\limsup_{p\to\infty}\lim_{n\to\infty}\sup_{u,v}\mathbb{E}\left|\sum_{j=1}^{N_p}\eta_j^{n,p}(u,v)+\sum_{i=N_p(p+1)+1}^{n-1}\xi_i^{(2)}(u,v)\right|^2$$
$$\le C\limsup_{p\to\infty}\lim_{n\to\infty}\left(\frac{1}{p+1}+p\Delta_n\right)=0.$$

For the term $\sum_{j=1}^{N_p}\zeta_j^{n,p}(u,v)$, we consider $\sum_{j=1}^{N_p}\zeta_j^{n,p\prime}(u,v)$ with

$$\zeta_j^{n,p\prime}(u,v)=\sum_{i=\alpha_j^p+1}^{\alpha_{j+1}^p-1}\xi_i^{(2)\prime}(u,v),$$

$$\xi_i^{(2)\prime}(u,v)=\Delta_n^{\frac12}\left\{\cos\left(\frac{\sqrt{2u}\sigma_{\alpha_j^p}^X\Delta_i^nW^X+\sqrt{2v}\sigma_{\alpha_j^p}^Y\Delta_{i+1}^nW^Y}{\sqrt{\Delta_n}}\right)-\mathrm{e}^{-\langle(u,v),((\sigma_{\alpha_j^p}^X)^2,(\sigma_{\alpha_j^p}^Y)^2)\rangle}\right\}.$$

Under assumption 2.2, we can show that

$$\sum_{j=1}^{N_p}\left(\zeta_j^{n,p}(u,v)-\zeta_j^{n,p\prime}(u,v)\right)\xrightarrow{P}0. \tag{A24}$$

The proof of the convergence in (A24) is similar to that in (A23). Denote $\mathcal{G}_j:=\mathcal{F}_{\alpha_j^p}$, then $\{\zeta_j^{n,p\prime}(u,v),j=1,2,\ldots,N_p\}$ is a $\mathcal{G}$-martingale sequence. To apply Theorem IX.7.28 of Jacod and Shiryaev (2003), it is sufficient to show the following convergence results:

$$\sum_{j=1}^{N_p}\mathbb{E}[\zeta_j^{n,p\prime}(u,v)\zeta_j^{n,p\prime}(u',v')|\mathcal{G}_j]$$
$$\xrightarrow{P}\int_0^T F(\sqrt{u}\sigma_s^X,\sqrt{v}\sigma_s^Y,\sqrt{u'}\sigma_s^X,\sqrt{v'}\sigma_s^Y,\rho_s)\mathrm{d}s, \tag{A25}$$

$$\sum_{j=1}^{N_p}\mathbb{E}[\left(\zeta_j^{n,p\prime}(u,v)\right)^4|\mathcal{G}_j]\xrightarrow{P}0, \tag{A26}$$

$$\sum_{j=1}^{N_p}\mathbb{E}[\zeta_j^{n,p\prime}(u,v)\Delta_jW^Z|\mathcal{G}_j]\xrightarrow{P}0,\quad\text{for } Z=X,Y, \tag{A27}$$

$$\sum_{j=1}^{N_p}\mathbb{E}[\zeta_j^{n,p\prime}(u,v)\Delta_jN|\mathcal{G}_j]\xrightarrow{P}0, \tag{A28}$$

$$\lim_{p\to\infty}F_p(\sqrt{u}\sigma_s^X,\sqrt{v}\sigma_s^Y,\sqrt{u'}\sigma_s^X,\sqrt{v'}\sigma_s^Y,\rho_s)$$
$$=F(\sqrt{u}\sigma_s^X,\sqrt{v}\sigma_s^Y,\sqrt{u'}\sigma_s^X,\sqrt{v'}\sigma_s^Y,\rho_s),$$

where N is any bounded $\mathcal{G}$-martingale orthogonal to W^X and W^Y,

$$F_p(\sqrt{u}\sigma_s^X,\sqrt{v}\sigma_s^Y,\sqrt{u'}\sigma_s^X,\sqrt{v'}\sigma_s^Y,\rho_s)$$
$$=\frac{p}{p+1}F_1(\sqrt{u}\sigma_s^X,\sqrt{v}\sigma_s^Y,\sqrt{u'}\sigma_s^X,\sqrt{v'}\sigma_s^Y,\rho_s)$$
$$+\frac{p-1}{p+1}F_2(\sqrt{u}\sigma_s^X,\sqrt{v}\sigma_s^Y,\sqrt{u'}\sigma_s^X,\sqrt{v'}\sigma_s^Y,\rho_s)$$
$$+\frac{p-1}{p+1}F_3(\sqrt{u}\sigma_s^X,\sqrt{v}\sigma_s^Y,\sqrt{u'}\sigma_s^X,\sqrt{v'}\sigma_s^Y,\rho_s),$$

and the definition of F_1, F_2 and F_3 will be given in the following proof. Note that the proofs of (A26)–(A28) are similar to those of (A16) to (A18). Thus we only give the proof of (A25). To show (A25), note that

$$\mathbb{E}[\zeta_j^{n,p\prime}(u,v)\zeta_j^{n,p\prime}(u',v')|\mathcal{G}_j]=\sum_{i=\alpha_j^p+1}^{\alpha_{j+1}^p-1}\mathbb{E}[\xi_i^{(2)\prime}(u,v)\xi_i^{(2)\prime}(u',v')|\mathcal{G}_j]$$
$$+\sum_{i=\alpha_j^p+1}^{\alpha_{j+1}^p-2}\mathbb{E}[\xi_i^{(2)\prime}(u,v)\xi_{i+1}^{(2)\prime}(u',v')|\mathcal{G}_j]$$
$$+\sum_{i=\alpha_j^p+1}^{\alpha_{j+1}^p-2}\mathbb{E}[\xi_i^{(2)\prime}(u',v')\xi_{i+1}^{(2)\prime}(u,v)|\mathcal{G}_j].$$

First,

$$\sum_{i=\alpha_j^p+1}^{\alpha_{j+1}^p-1}\mathbb{E}[\xi_i^{(2)\prime}(u,v)\xi_i^{(2)\prime}(u',v')|\mathcal{G}_j]$$
$$=\frac{p\Delta_n}{2}\mathrm{e}^{-u(\sigma_{\alpha_j^p}^X)^2-v(\sigma_{\alpha_j^p}^Y)^2-u'(\sigma_{\alpha_j^p}^X)^2-v'(\sigma_{\alpha_j^p}^Y)^2}$$
$$\times\{\mathrm{e}^{-2((\sqrt{uu'}\sigma_{\alpha_j^p}^X)^2+(\sqrt{vv'}\sigma_{\alpha_j^p}^Y)^2)}+\mathrm{e}^{2((\sqrt{uu'}\sigma_{\alpha_j^p}^X)^2+(\sqrt{vv'}\sigma_{\alpha_j^p}^Y)^2)}-2\}$$
$$=:\frac{p\Delta_n}{2}F_1(\sqrt{u}\sigma_{\alpha_j^p}^X,\sqrt{v}\sigma_{\alpha_j^p}^Y,\sqrt{u'}\sigma_{\alpha_j^p}^X,\sqrt{v'}\sigma_{\alpha_j^p}^Y,\rho_{\alpha_j^p}).$$

Second,

$$\sum_{i=\alpha_j^p+1}^{\alpha_{j+1}^p-2}\mathbb{E}[\xi_i^{(2)\prime}(u,v)\xi_{i+1}^{(2)\prime}(u',v')|\mathcal{G}_j]$$
$$=\frac{(p-1)\Delta_n}{2}\mathrm{e}^{-u(\sigma_{\alpha_j^p}^X)^2-v(\sigma_{\alpha_j^p}^Y)^2-u'(\sigma_{\alpha_j^p}^X)^2-v'(\sigma_{\alpha_j^p}^Y)^2}$$
$$\times\{\mathrm{e}^{-2\sqrt{u'v}\sigma_{\alpha_j^p}^X\sigma_{\alpha_j^p}^Y\rho_{\alpha_j^p}}+\mathrm{e}^{2\sqrt{u'v}\sigma_{\alpha_j^p}^X\sigma_{\alpha_j^p}^Y\rho_{\alpha_j^p}}-2\}$$

$$=: \frac{(p-1)\Delta_n}{2} F_2(\sqrt{u}\sigma^X_{\alpha^p_j}, \sqrt{v}\sigma^Y_{\alpha^p_j}, \sqrt{u'}\sigma^X_{\alpha^p_j}, \sqrt{v'}\sigma^Y_{\alpha^p_j}, \rho_{\alpha^p_j}).$$

Third,

$$\sum_{i=\alpha^p_j+1}^{\alpha^p_{j+1}-2} \mathbb{E}[\xi^{(2)\prime}_i(u',v')\xi^{(2)\prime}_{i+1}(u,v)|\mathcal{G}_j]$$
$$= \frac{(p-1)\Delta_n}{2} \mathrm{e}^{-u(\sigma^X_{\alpha^p_j})^2 - v(\sigma^Y_{\alpha^p_j})^2 - u'(\sigma^X_{\alpha^p_j})^2 - v'(\sigma^Y_{\alpha^p_j})^2}$$
$$\times \{\mathrm{e}^{-2\sqrt{uv'}\sigma^X_{\alpha^p_j}\sigma^Y_{\alpha^p_j}\rho_{\alpha^p_j}} + \mathrm{e}^{2\sqrt{uv'}\sigma^X_{\alpha^p_j}\sigma^Y_{\alpha^p_j}\rho_{\alpha^p_j}} - 2\}$$
$$=: \frac{(p-1)\Delta_n}{2} F_3(\sqrt{u}\sigma^X_{\alpha^p_j}, \sqrt{v}\sigma^Y_{\alpha^p_j}, \sqrt{u'}\sigma^X_{\alpha^p_j}, \sqrt{v'}\sigma^Y_{\alpha^p_j}, \rho_{\alpha^p_j}).$$

Then we can obtain (A25). Next the tightness argument in theorem 3.1 also proves the tightness of $\sum_{i=1}^{n-1}\sum_{j=1}^{3}\xi^{(j)}_i(u,v)$. Hence, the proof of the convergence in (5) is complete.

Finally, the proof of the consistency of $\tilde{\Gamma}_n$ is showed as follows:

$$\tilde{\Gamma}_n = \Delta_n \sum_{i=1}^{n-1} \cos\left(\frac{\sqrt{2u}\Delta^n_i X + \sqrt{2v}\Delta^n_{i+1}Y}{\sqrt{\Delta_n}}\right)$$
$$\times \cos\left(\frac{\sqrt{2u'}\Delta^n_i X + \sqrt{2v'}\Delta^n_{i+1}Y}{\sqrt{\Delta_n}}\right)$$
$$+ \Delta_n \sum_{i=1}^{n-2} \cos\left(\frac{\sqrt{2u}\Delta^n_{i-1}X}{\sqrt{\Delta_n}}\right)\cos\left(\frac{\sqrt{2v}\Delta^n_{i+1}Y}{\sqrt{\Delta_n}}\right)$$
$$\times \cos\left(\frac{\sqrt{2u'}\Delta^n_{i+1}X}{\sqrt{\Delta_n}}\right)\cos\left(\frac{\sqrt{2v'}\Delta^n_{i+2}Y}{\sqrt{\Delta_n}}\right)$$
$$+ \Delta_n \sum_{i=1}^{n-2} \cos\left(\frac{\sqrt{2u'}\Delta^n_{i-1}X}{\sqrt{\Delta_n}}\right)\cos\left(\frac{\sqrt{2v'}\Delta^n_{i+1}Y}{\sqrt{\Delta_n}}\right)$$
$$\times \cos\left(\frac{\sqrt{2u}\Delta^n_{i+1}X}{\sqrt{\Delta_n}}\right)\cos\left(\frac{\sqrt{2v}\Delta^n_{i+2}Y}{\sqrt{\Delta_n}}\right)$$
$$- 3\Delta_n \sum_{i=1}^{n-1} \cos\left(\frac{\sqrt{2(u+u')}\Delta^n_i X + \sqrt{2(v+v')}\Delta^n_{i+1}Y}{\sqrt{\Delta_n}}\right)$$
$$= \Delta_n \sum_{i=1}^{n-1} \cos\left(\frac{\sqrt{2u}\Delta^n_i X + \sqrt{2v}\Delta^n_{i+1}Y}{\sqrt{\Delta_n}}\right)$$
$$\times \cos\left(\frac{\sqrt{2u'}\Delta^n_i X + \sqrt{2v'}\Delta^n_{i+1}Y}{\sqrt{\Delta_n}}\right)$$
$$- \Delta_n \sum_{i=1}^{n-1} \cos\left(\frac{\sqrt{2(u+u')}\Delta^n_i X + \sqrt{2(v+v')}\Delta^n_{i+1}Y}{\sqrt{\Delta_n}}\right)$$
$$+ \Delta_n \sum_{i=1}^{n-2} \cos\left(\frac{\sqrt{2u}\Delta^n_{i-1}X}{\sqrt{\Delta_n}}\right)\cos\left(\frac{\sqrt{2v}\Delta^n_{i+1}Y}{\sqrt{\Delta_n}}\right)$$
$$\times \cos\left(\frac{\sqrt{2u'}\Delta^n_{i+1}X}{\sqrt{\Delta_n}}\right)\cos\left(\frac{\sqrt{2v'}\Delta^n_{i+2}Y}{\sqrt{\Delta_n}}\right)$$
$$- \Delta_n \sum_{i=1}^{n-1} \cos\left(\frac{\sqrt{2(u+u')}\Delta^n_i X + \sqrt{2(v+v')}\Delta^n_{i+1}Y}{\sqrt{\Delta_n}}\right)$$
$$+ \Delta_n \sum_{i=1}^{n-2} \cos\left(\frac{\sqrt{2u'}\Delta^n_{i-1}X}{\sqrt{\Delta_n}}\right)\cos\left(\frac{\sqrt{2v'}\Delta^n_{i+1}Y}{\sqrt{\Delta_n}}\right)$$
$$\times \cos\left(\frac{\sqrt{2u}\Delta^n_{i+1}X}{\sqrt{\Delta_n}}\right)\cos\left(\frac{\sqrt{2v}\Delta^n_{i+2}Y}{\sqrt{\Delta_n}}\right)$$
$$- \Delta_n \sum_{i=1}^{n-1} \cos\left(\frac{\sqrt{2(u+u')}\Delta^n_i X + \sqrt{2(v+v')}\Delta^n_{i+1}Y}{\sqrt{\Delta_n}}\right).$$

Because of the Law of Large Numbers

$$\tilde{\Gamma}_n - \Bigg[\frac{1}{2}\Delta_n \sum_{i=1}^{n-1} \mathrm{e}^{-u(\sigma^X_{i-1})^2 - v(\sigma^X_{i-1})^2 - u'(\sigma^Y_{i-1})^2 - v'(\sigma^Y_{i-1})^2}$$
$$\times \left[\mathrm{e}^{-2\sqrt{uu'}(\sigma^X_{i-1})^2 + 2\sqrt{uu'}(\sigma^X_{i-1})^2 - 2\sqrt{vv'}(\sigma^Y_{i-1})^2 + 2\sqrt{vv'}(\sigma^Y_{i-1})^2} - 2\right]$$
$$+ \frac{1}{2}\Delta_n \sum_{i=1}^{n-1} \mathrm{e}^{-u(\sigma^X_{i-1})^2 - v(\sigma^X_{i-1})^2 - u'(\sigma^Y_{i-1})^2 - v'(\sigma^Y_{i-1})^2}$$
$$\times \left[\mathrm{e}^{-2\sqrt{vu'}\rho\sigma^X_{i-1}\sigma^Y_{i-1} + 2\sqrt{vu'}\rho\sigma^X_{i-1}\sigma^Y_{i-1}} - 2\right]$$
$$+ \frac{1}{2}\Delta_n \sum_{i=1}^{n-1} \mathrm{e}^{-u(\sigma^X_{i-1})^2 - v(\sigma^X_{i-1})^2 - u'(\sigma^Y_{i-1})^2 - v'(\sigma^Y_{i-1})^2}$$
$$\times \left[\mathrm{e}^{-2\sqrt{v'u}\rho\sigma^X_{i-1}\sigma^Y_{i-1} + 2\sqrt{v'u}\rho\sigma^X_{i-1}\sigma^Y_{i-1}} - 2\right]\Bigg] \xrightarrow{P} 0$$

and

$$\frac{1}{2}\Delta_n \sum_{i=1}^{n-1} \mathrm{e}^{-u(\sigma^X_{i-1})^2 - v(\sigma^X_{i-1})^2 - u'(\sigma^Y_{i-1})^2 - v'(\sigma^Y_{i-1})^2}$$
$$\times \left[\mathrm{e}^{-2\sqrt{uu'}(\sigma^X_{i-1})^2 + 2\sqrt{uu'}(\sigma^X_{i-1})^2 - 2\sqrt{vv'}(\sigma^Y_{i-1})^2 + 2\sqrt{vv'}(\sigma^Y_{i-1})^2} - 2\right]$$
$$+ \frac{1}{2}\Delta_n \sum_{i=1}^{n-1} \mathrm{e}^{-u(\sigma^X_{i-1})^2 - v(\sigma^X_{i-1})^2 - u'(\sigma^Y_{i-1})^2 - v'(\sigma^Y_{i-1})^2}$$
$$\left[\mathrm{e}^{-2\sqrt{vu'}\rho\sigma^X_{i-1}\sigma^Y_{i-1} + 2\sqrt{vu'}\rho\sigma^X_{i-1}\sigma^Y_{i-1}} - 2\right]$$
$$+ \frac{1}{2}\Delta_n \sum_{i=1}^{n-1} \mathrm{e}^{-u(\sigma^X_{i-1})^2 - v(\sigma^X_{i-1})^2 - u'(\sigma^Y_{i-1})^2 - v'(\sigma^Y_{i-1})^2}$$
$$\times \left[\mathrm{e}^{-2\sqrt{v'u}\rho\sigma^X_{i-1}\sigma^Y_{i-1} + 2\sqrt{v'u}\rho\sigma^X_{i-1}\sigma^Y_{i-1}} - 2\right]$$
$$\xrightarrow{P} \frac{1}{2}\int_0^T \mathrm{e}^{-u(\sigma^X_s)^2 - v(\sigma^X_s)^2 - u'(\sigma^Y_s)^2 - v'(\sigma^Y_s)^2}$$
$$\times \Big[\mathrm{e}^{-2\sqrt{uu'}(\sigma^X_s)^2 + 2\sqrt{uu'}(\sigma^X_s)^2 - 2\sqrt{vv'}(\sigma^Y_s)^2 + 2\sqrt{vv'}(\sigma^Y_2)^2}$$
$$+ \mathrm{e}^{-2\sqrt{vu'}\rho\sigma^X_s\sigma^Y_s + 2\sqrt{vu'}\rho\sigma^X_s\sigma^Y_s}$$
$$+ \mathrm{e}^{-2\sqrt{v'u}\rho\sigma^X_s\sigma^Y_s + 2\sqrt{v'u}\rho\sigma^X_s\sigma^Y_s} - 6\Big]\mathrm{d}s.$$

We can achieve

$$\tilde{\Gamma}_n \xrightarrow{P} \frac{1}{2}\int_0^T \mathrm{e}^{-u(\sigma^X_s)^2 - v(\sigma^X_s)^2 - u'(\sigma^Y_s)^2 - v'(\sigma^Y_s)^2}$$
$$\times \Big[\mathrm{e}^{-2\sqrt{uu'}(\sigma^X_s)^2 + 2\sqrt{uu'}(\sigma^X_s)^2 - 2\sqrt{vv'}(\sigma^Y_s)^2 + 2\sqrt{vv'}(\sigma^Y_2)^2}$$
$$+ \mathrm{e}^{-2\sqrt{vu'}\rho\sigma^X_s\sigma^Y_s + 2\sqrt{vu'}\rho\sigma^X_s\sigma^Y_s}$$
$$+ \mathrm{e}^{-2\sqrt{v'u}\rho\sigma^X_s\sigma^Y_s + 2\sqrt{v'u}\rho\sigma^X_s\sigma^Y_s} - 6\Big]\mathrm{d}s.$$

■

Proof of theorem 4.3. (i) For

$$\sqrt{T}\begin{pmatrix} \frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^{x,y}(u,v)-\frac{1}{T}\sum_{t=1}^{T}Z_t^{x,y}(u,v) \\ +\frac{1}{T}\sum_{t=1}^{T}Z_t^{x,y}(u,v)-\mu_t^{x,y}(u,v) \\ \frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^{x}(u)-\frac{1}{T}\sum_{t=1}^{T}Z_t^{x}(u)+\frac{1}{T}\sum_{t=1}^{T}Z_t^{x}(u)-\mu_t^{x}(u) \\ \frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^{y}(v)-\frac{1}{T}\sum_{t=1}^{T}Z_t^{y}(v)+\frac{1}{T}\sum_{t=1}^{T}Z_t^{y}(v)-\mu_t^{y}(v) \end{pmatrix}$$

Combing with assumptions 4.1, 4.2, theorem 3.2, we have

$$\sqrt{T}\left(\frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^{x,y}(u,v)-\frac{1}{T}\sum_{t=1}^{T}Z_t^{x,y}(u,v)\right)=O_p(\sqrt{T\Delta_n}),$$

$$\frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^{x}(u)-\frac{1}{T}\sum_{t=1}^{T}Z_t^{x}(u)=O_p(\sqrt{T\Delta_n})$$

$$\frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^{y}(v)-\frac{1}{T}\sum_{t=1}^{T}Z_t^{y}(v)=O_p(\sqrt{T\Delta_n}).$$

Then according to assumption 4.1 and CLT for stationary processes by theorem VIII.3.79 of Jacod and Shiryaev (2003), the finite-dimensional convergence is as follows:

$$\sqrt{T}\begin{pmatrix} \frac{1}{T}\sum_{t=1}^{T}Z_t^{x,y}(u,v)-\mu_t^{x,y}(u,v) \\ \frac{1}{T}\sum_{t=1}^{T}Z_t^{x}(u)-\mu_t^{x}(u) \\ \frac{1}{T}\sum_{t=1}^{T}Z_t^{y}(v)-\mu_t^{y}(v) \end{pmatrix}\xrightarrow{f-d}\Phi(u,v),$$

where $\Phi(u,v)$ is shown in theorem 4.3.

Next, we show the tightness. For $u_1,u_2,v_1,v_2\in\mathbb{R}_+$,

$$\begin{aligned}
&T\mathbb{E}\Bigg[\frac{1}{T}\sum_{t=1}^{T}Z_t^{x,y}(u_1,v_1)-\mu_t^{x,y}(u_1,v_1) \\
&\quad -\frac{1}{T}\sum_{t=1}^{T}Z_t^{x,y}(u_2,v_2)+\mu_t^{x,y}(u_2,v_2)\Bigg]^2 \\
&=\frac{1}{T}\mathbb{E}\Bigg[\sum_{t=1}^{T}\Bigg[\left(\int_{t-1}^{t}e^{-u_1(\sigma_s^x)^2-v_1(\sigma_s^y)^2}-e^{-u_2(\sigma_s^x)^2-v_2(\sigma_s^y)^2}\mathrm{d}s\right) \\
&\quad -\mathbb{E}\left(\int_{t-1}^{t}e^{-u_1(\sigma_s^x)^2-v_1(\sigma_s^y)^2}-e^{-u_2(\sigma_s^x)^2-v_2(\sigma_s^y)^2}\mathrm{d}s\right)\Bigg]\Bigg]^2 \\
&\le C[(u_1-u_2)^2+(v_1-v_2)^2]
\end{aligned}$$

The last inequality follows from assumption 4.1 and Taylor's expansion,

$$\begin{aligned}
&e^{-u_1(\sigma_s^x)^2-v_1(\sigma_s^y)^2}-e^{-u_2(\sigma_s^x)^2-v_2(\sigma_s^y)^2} \\
&=\left(-(\sigma_{s^\star}^x)^2e^{-u_2(\sigma_{s^\star}^x)^2-v_2(\sigma_{s^\star}^y)^2}\right)(u_1-u_2) \\
&\quad+\left(-(\sigma_{s^\star}^y)^2e^{-u_2(\sigma_{s^\star}^x)^2-v_2(\sigma_{s^\star}^y)^2}\right)(v_1-v_2)
\end{aligned}$$

where $s^\star\in(t-1,t)$. The desired results can be obtained from theorem 20 of Ibragimov and Has' Minskii (1981).

(ii) We denote

$$C_{11}^l([u,v],[u',v'])=\frac{1}{T}\sum_{t=l+1}^{T}[Z_t^{x,y}(u,v)-\mu_t^{x,y}(u,v)] [Z_{t-l}^{x,y}(u',v')-\mu_t^{x,y}(u',v')].$$

Based on proposition 1 in Andrews (1991),

$$C_{11}^0([u,v],[u',v'])+\sum_{i=1}^{L_T}\omega(i,L_T)(C_{11}^i([u,v],[u',v']) + C_{11}^i([u',v'],[u,v]))\xrightarrow{P}V_{11}([u,v],[u',v'])$$

And also $Z_t^{x,y}(u,v)$, $\hat{Z}_t^{x,y}(u,v)$ are bounded,

$$\begin{aligned}
&|\hat{C}_{11}^i-C_{11}^i| \\
&\le C\frac{1}{T}\sum_{t=1}^{T}\left\{\left|\hat{Z}_t^{x,y}(u,v)-Z_t^{x,y}(u,v)\right|\right. \\
&\quad\left.+\left|\frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^{x,y}(u,v)-\mu_t^{x,y}(u,v)\right|\right\} \\
&\le C\frac{1}{T}\sum_{t=1}^{T}\left\{\left|\hat{Z}_t^{x,y}(u,v)-Z_t^{x,y}(u,v)\right|\right. \\
&\quad+\left|\frac{1}{T}\sum_{t=1}^{T}\hat{Z}_t^{x,y}(u,v)-\frac{1}{T}\sum_{t=1}^{T}Z_t^{x,y}(u,v)\right| \\
&\quad\left.+\left|\frac{1}{T}\sum_{t=1}^{T}Z_t^{x,y}(u,v)-\mu_t^{x,y}(u,v)\right|\right\}
\end{aligned}\tag{A30}$$

The first and second terms of (A30) are $O_p(\sqrt{\Delta_n})$ according to theorem 3.2, the magnitude of the last term is $O_p(\frac{1}{\sqrt{T}})$ due to theorem 4.3. Therefore, the relationship between T,Δ_n and L_T can be obtained from $\sqrt{\Delta_n L_T}\to 0, L_T\frac{1}{\sqrt{T}}\to 0$. Then we use the same way to achieve the consistency of $\hat{V}_{ab}^i$, $a=1,2,3$, $b=1,2,3$. ■

Proof of proposition 4.5. Under H_0, we have $\mathbb{P}(||\hat{\mathcal{S}}_{T,n}||^2>d_\alpha|H_0)\to\alpha$. Under H_1, we can see the test statistic $||\hat{\mathcal{S}}_{T,n}||^2$ goes to infinity when $T\to\infty$ and $\Delta_n\to 0$, therefore, $\mathbb{P}(||\hat{\mathcal{S}}_{T,n}||^2>d_\alpha|H_1)\to 1$. In addition, $\hat{d}_\alpha$ is a consistent estimator of d_α, which proves the desired result. ■